Can Physical Forces Find a Magic Square? Spectral Energy Landscapes, Window Entropies, and Interactive Relaxation Dynamics

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Abstract

If we attach virtual springs to the rows, columns, and diagonals of a scrambled $n \times n$ grid of numbers $1 \dots n^2$—or let the numbers float as buoyant "bubbles" in a two-dimensional dish driven by line-sum imbalances—can physical relaxation discover a magic square on its own? Whereas the first three papers of the *Magic Gems* series studied the static geometry (Paper I), polynomial moment identities (Paper II), and rigid-body rotational mechanics (Paper III) of completed magic squares, this paper and its open-source interactive browser laboratory (`physicalSolver`) investigate the inverse dynamical landscape governed by the quadratic line-sum Hamiltonian $\mathcal{H}_{\text{line}}(S) = \sum_{L \in \mathcal{L}_n} R_L(S)^2 = \mathbf{z}^\top L_n \mathbf{z}$ on the symmetric group S_{n^2}. Combining elementary linear algebra with exhaustive computer censuses and interactive simulations, we uncover four surprises across scales. First, we show why balancing a square as a flat 3D plate with zero gravitational torque ($\langle X, Z\rangle = \langle Y, Z\rangle = 0$) and balanced main diagonals ($R_{D_\pm} = 0$) fails to produce a magic square: restricting the Rook's-graph-plus-diagonals operator $L_n = A_n^\top A_n$ to the zero-mean hyperplane $\mathbf{1}^\perp$ collapses its irrational eigenvalues into exact integers with universal average $\bar{\mu}(\widetilde{L}_n) = 2$ and random-shuffle expectation $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = \frac{2}{3}n^2(n^4-1)$ (Proposition 2.1), but tilt and diagonal balance fix only 4 modes out of $2n$, leaving $2n-4$ alternating row and column modes unconstrained—vividly illustrated by an explicit 4×4 "impostor square" S_{low} at $\mathcal{H}_{\text{line}} = 1{,}584$ (Remark 2.2). Second, inside the ground-state manifold $\mathcal{H}_{\text{line}}(S) = 0$ where all 1D line sums are equal, we use sliding $k \times k$ window entropies $\widetilde{H}_p(k)$ and a Concentric Shell Quadrupole Identity $M_{xyz}(S) = -8\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S)$ (Lemma 3.1 and Proposition 3.2) to map all 880 order-4 magic squares, explaining both the 48 symmetric associative squares and 8 unexpected non-symmetric squares in Dudeney Groups IX and X whose corner and edge shells cancel in an exact 3 : 8 coprime resonance. Third, by mapping all $9! = 362{,}880$ permutations of S_9 (Theorem 4.2), we prove that greedy 2-swap descent converges to the Lo Shu square from 74.53% of all starting shuffles and gets stuck from the remaining 25.47% in a *single* dihedral trap orbit $S^{(3)}_{\text{trap}}$ at $\mathcal{H}_{\text{line}} = 8$; no 2-swap and no 3-cycle can improve $S^{(3)}_{\text{trap}}$, but *one and only one* of the 378 disjoint $(2,2)$ double swaps—the row-preserving move $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$, acting like a 4-cell Kempe chain—unlocks the trap in a single step. Fourth, in our continuous *Bubble Trinity* simulator, we show why soft 2D Gaussian particles lower their field energy by 37.5% yet fail (0%) under both greedy snapping and exact Hungarian bipartite matching by huddling between grid cells to average out heights, whereas a discrete *Multi-Exchange Iterated Local Search (ME-ILS)* engine achieves 100.0% synthesis from $n = 3$ to $n = 10$ (up to $100! \approx 9.33 \times 10^{157}$ states) and reveals that $n = 5$ is the hardest frustration bottleneck (8.03 restarts) before higher orders ($n = 6 \dots 10$, dropping to 3.34 restarts at $n = 10$) gain "dimensional relief" as $1 - 2/(n-1)^2$ of the $(2,2)$ plaquette moves become diagonal-free.

1 Introduction: From Static Magic Gems to Interactive Physical Relaxation

For more than two millennia—from the 3×3 Lo Shu turtle diagram of ancient China to Albrecht Dürer's 1514 engraving *Melencolia I* and Bernard Frénicle de Bessy's 1693 enumeration of all

880 order-4 squares—magic squares have fascinated mathematicians, artists, and physicists alike [11, 8, 10, 1]. A magic square of order n is an arrangement $S : \{1, \dots, n\}^2 \to \{1, \dots, n^2\}$ of the consecutive integers $1 \dots n^2$ in an $n \times n$ grid such that every row, every column, and both main diagonals sum to the magic constant $M_n = \frac{1}{2}n(n^2 + 1)$. When we lift each grid cell into three-dimensional space using centred integer coordinates

$$X_j = 2j - (n + 1), \qquad Y_i = (n + 1) - 2i, \qquad Z_{ij} = 2s_{ij} - (n^2 + 1), \tag{1}$$

each arrangement $S \in S_{n^2}$ becomes a 3D cloud of n^2 points $\mathcal{P}(S) = \{(X_j, Y_i, Z_{ij})\} \subset \mathbb{Z}^3$ whose convex hull $K(S) = \operatorname{conv} \mathcal{P}(S)$ is a polyhedral sculpture that we call a *Magic Gem*. Across the first three papers of this series, we explored the geometry and mechanics of these sculptures once they are already built. In Paper I [20], we classified the polyhedral faces, vertex counts, surface areas, volumes, and symmetry groups of Magic Gems across all 880 order-4 squares and higher-order families. In Paper II [19], we showed that equal row and column sums force the bivariate polynomial moments $\langle X^j, Z\rangle = \langle Y^j, Z\rangle = 0$ to vanish for all $j \geq 0$, which automatically diagonalizes the 3×3 moment-of-inertia tensor of every magic square [16, 26]. In Paper III [18], we put those algebraic identities to a physical test, showing that every equal-vertex-mass Magic Gem spins in 3D as an oblate symmetric top ($I_{xx} = I_{yy} < I_{zz}$) immune to intermediate-axis Dzhanibekov tumbling, while solid 3D-printed polyhedra remain statically stable on every single one of their 50,334 triangular facets.

Those three studies, however, only examined *completed* magic squares. When demonstrating the 3D Magic Gem viewer to students and colleagues, a natural hands-on question kept coming up: *Instead of typing in a known magic square, what happens if we start with a randomly scrambled grid of numbers $1 \dots n^2$, attach virtual physical forces that pull every row, column, and diagonal toward the magic sum M_n, and watch the system try to settle into a magic square on its own?* Of course, if our only goal were to write down a single magic square of order n on a computer, we would not need a physical simulation at all: classical closed-form recipes—such as de La Loubère's diagonal staircase (Siamese) rule for odd n, complementary X-masks for doubly-even $n = 4k$, and Conway's LUX method for singly-even $n = 4k+2$—construct a solution in $O(n^2)$ steps [8, 1, 2], and specialized constraint-based local search (CBLS) algorithms [7, 28] can assemble magic squares of order $n \sim 100$ by targeting defective lines directly. Our question is different and much closer in spirit to experimental mathematics, statistical physics, and interactive visualization [23, 29, 14]: *What does the energy landscape of magic-square constraints actually look like across the $n^2!$ permutations of S_{n^2}, why do intuitive physical relaxation schemes get stuck in deceptive non-magic traps, and what cooperative moves unlock them?*

To explore this question visually and interactively, we built an open-source browser laboratory, `physicalSolver` [21], organized around four interactive web applications: an *Energy Decomposition Explorer* (`energy-decomposition.html`) that splits any square's error into 3D tilt, line residuals, sliding $k \times k$ sub-window entropies, and complement-pair symmetry breaks; a *Discrete Physics Lab* (`physics-lab.html`) and *Vector-Physics Explorer* (`vector-physics.html`) that visualize cell-by-cell stress vectors and test 2-swap and 3-cycle descent; and a continuous particle sandbox called the *Bubble Trinity* (`bubble-trinity.html`), in which the n^2 numbers float in a 2D dish as buoyant, size-modulated bubbles driven by line-sum imbalances, core repulsion, and home-cell wells. Playing with these four interactive tools immediately revealed a series of counterintuitive behaviors—such as balanced plates that refuse to become magic, a stubborn 3×3 configuration that traps a quarter of all random shuffles, floating bubbles that lower their field energy on screen yet fail to snap into a valid square, and 6×6 or 10×10 grids that untangle with fewer restarts than 5×5 grids.

Following Whitesides's blueprint-driven approach to connecting interactive experiments with mathematical structure [32], this paper provides the complete mathematical and computational explanation for what happens inside `physicalSolver`. Let $\mathcal{L}_n$ denote the set of $2n+2$ magic lines (n rows, n columns, and 2 main diagonals), and let $R_L(S) = \sum_{(i,j)\in L} Z_{ij} = 2\big(\sum_{(i,j)\in L} s_{ij} - M_n\big)$ be the centred residual of line $L \in \mathcal{L}_n$. Stacking the $2n+2$ line indicators as rows of the incidence

Table 1: The four interactive laboratories of `physicalSolver` and the corresponding mathematical results established in Paper D.

Section & Web Lab	Interactive Phenomenon / Question	Mathematical Explanation / Census	Headline Invariant
Sec. 2: Spectral & Tilt Lab	Why does a 3D-balanced plate with magic diagonals fail to be magic?	Proposition 2.1 ($\mathbf{1}^\perp$ integer spectrum) Remark 2.2 ($2n-4$ hidden modes)	$\mathrm{spec}(\widetilde{L}_n) \subset \mathbb{Z}$, $\bar{\mu}(\widetilde{L}_n) = 2$ Impostor S_{low}: $E_{\text{low}} = 0$, $\mathcal{H}_{\text{line}} = 1{,}584$
Sec. 3: Window & Texture Lab	How do the 880 4×4 magic squares differ in 2D texture and 3D twist?	Lemma 3.1 (interior $\Leftrightarrow$ toroidal 2×2) Proposition 3.2 ($M_{xyz} = -8Q_{\text{corner}} - 3Q_{\text{edge}}$)	$W^{\text{int}}_{2\times2} = 0 \iff$ Group I (48/880) 56 octupole zeros (48 Grp III +8 Grp IX/X)
Sec. 4: Discrete Swap Lab	Why do pair and triplet swaps get stuck 25.47% of the time at $n = 3$?	Lemma 4.1 & Theorem 4.2 (362,880 S_9 map) Unique Kempe-like $(2,2)$ double swap	1 trap orbit $S^{(3)}_{\text{trap}}$ captures 25.47% of S_9 $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$ restores 100% basin
Sec. 5: Bubble Trinity Lab	Why do continuous 2D bubbles cheat, and why is $n = 5$ harder than $n = 10$?	Regimes I–IV (greedy & Hungarian snap) Multi-Exchange ILS ($n = 3\ldots10$, 900 runs)	$\mathcal{H}_{\text{ind}}$ drops 37.5%, discrete magic = 0% 100% solved; restarts peak at $n = 5$ (8.03)

matrix $A_n \in \{0,1\}^{(2n+2)\times n^2}$, the *line-indicator Hamiltonian* is the sum of squared line residuals:

$$\mathcal{H}_{\text{line}}(S) = \sum_{L\in\mathcal{L}_n} R_L(S)^2 = \|A_n\mathbf{z}\|_2^2 = \mathbf{z}^\top L_n \mathbf{z}, \qquad L_n = A_n^\top A_n \in \mathbb{Z}^{n^2\times n^2}. \tag{2}$$

Because every centred residual $R_L(S)$ is twice the standard line error $\sum_{(i,j)\in L} s_{ij} - M_n$, $\mathcal{H}_{\text{line}}(S)$ is four times the standard sum of squared line errors ($E_{\text{std}} = \frac{1}{4}\mathcal{H}_{\text{line}}$), and $\mathcal{H}_{\text{line}}(S) = 0$ if and only if S is a magic square.

Overview of the Interactive Laboratories and Main Findings

Each section of this paper pairs one of the interactive views in `physicalSolver` with an exact algebraic analysis or exhaustive computer census (Table 1):

In Section 2, we start with the puzzle that opens the *Energy Decomposition Explorer* (`energy-decomposition.html`): if we adjust a square until its 3D center-of-mass tilt vanishes ($\langle X, Z\rangle = \langle Y, Z\rangle = 0$) and both main diagonals sum to M_n ($R_{D_+} = R_{D_-} = 0$), have we found a magic square? We exhibit a striking 4×4 counterexample S_{low} (Figure 1b) that balances flat with zero diagonal error yet has wildly unequal rows and columns ($\mathcal{H}_{\text{line}} = 1{,}584$). Using the Rook's-graph-plus-diagonals decomposition of $L_n = A_n^\top A_n$ [12, 4, 17], we show in Proposition 2.1 that restricting L_n to the zero-mean hyperplane $\mathbf{1}^\perp$ inhabited by all centred permutations turns its irrational eigenvalues $\lambda_\pm(n)$ into exact integers—$\{0^{[n^2-2n-1]}, (n-2)^{[1]}, n^{[2n-1]}\}$ for even n and $\{0^{[n^2-2n-1]}, (n-1)^{[2]}, n^{[2n-2]}\}$ for odd n—with a universal average eigenvalue $\bar{\mu}(\widetilde{L}_n) = 2$ and random-shuffle expectation $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$ (Figure 1a). This spectrum immediately reveals why S_{low} exists (Remark 2.2): linear tilt and diagonal sums only constrain 4 directions out of the $2n$-dimensional active subspace on $\mathbf{1}^\perp$, leaving $2n-4$ alternating row and column modes completely hidden from rigid-body balance.

In Section 3, we use the second panel of `energy-decomposition.html` to peer *inside* the magic ground-state manifold $\mathcal{H}_{\text{line}}(S) = 0$, where 1D line sums can no longer tell one magic square from another. Evaluating sliding $k\times k$ sub-window energies and Shannon entropies $\widetilde{H}_p(k)$ across classical constructions ($n = 3, 4, 5, 6, 8$) and all 880 order-4 squares (Figure 2), we show in Lemma 3.1 that on any even-order magic square ($n = 2m$), balancing the $(2m-1)^2$ interior 2×2 quartets automatically balances all boundary-wrapping toroidal quartets, isolating the 48 panmagic squares of Dudeney Group I at $n = 4$. We then prove the Concentric Shell Quadrupole Identity $M_{xyz}(S) = -8\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S)$ (Proposition 3.2), which explains both why the complement-break count $B_{\text{comp}}(S) \in \{0, 8\}$ is all-or-nothing at order 4 and why the 3D twist moment $M_{xyz}(S)$ vanishes on the 48 symmetric associative squares of Group III plus *exactly* 8 *non-symmetric squares* in Groups IX and X where the corner and edge shells cancel in a $(-3k, 8k)$ resonance (Figure 3).

In Section 4, we turn to the *Discrete Physics Lab* (`physics-lab.html`) and *Vector-Physics Explorer* (`vector-physics.html`), where users watch a scrambled square relax by swapping pairs of cells that lower $\mathcal{H}_{\text{line}}(S)$ according to the quadratic update $\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) = -2d(g_a - g_b) + c_{ab}d^2$ (Lemma 4.1). By mapping all $9! = 362{,}880$ permutations of S_9 (Theorem 4.2 and Figure 4), we prove that 74.53% of all 3×3 starting grids roll straight down to the Lo Shu magic square, while the remaining 25.47% (92,438 states) get stuck in a *single* dihedral trap orbit $S^{(3)}_{\text{trap}}$ at $\mathcal{H}_{\text{line}} = 8$. Like a

mechanical combination lock, $S_{\text{trap}}^{(3)}$ cannot be improved by any single 2-swap ($\min \Delta\mathcal{H}_{\text{line}} = +4$) or any 3-way rotation ($\min \Delta\mathcal{H}_{\text{line}} = +8$). Yet out of all 378 disjoint $(2,2)$ double swaps on the 3×3 board, *one and only one*—the simultaneous intra-row swap $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$, acting like a 4-vertex Kempe chain in graph coloring—unlocks $S_{\text{trap}}^{(3)}$ directly to $\mathcal{H}_{\text{line}} = 0$, expanding the deterministic basin of S_9 to 100.00%.

Finally, in Section 5, we analyze the flagship *Bubble Trinity* laboratory (`bubble-trinity.html`), contrasting continuous 2D floating-bubble relaxation against our discrete *Multi-Exchange Iterated Local Search (ME-ILS)* engine (Figure 6 and Table 7). Watching the continuous bubbles drift across the four interactive presets (*Frozen Lattice*, *Slow Drift*, *Field Storm*, and *Centrifugal Balance*) reveals a fascinating lesson in continuous vs. discrete optimization: the soft Gaussian field energy $\mathcal{H}_{\text{ind}}(t)$ drops by 37.5% on screen, yet snapping the bubbles to the grid—whether by greedy nearest-cell assignment or by exact $O(N^3)$ Hungarian bipartite matching—produces 0.0% magic squares, because soft bubbles "cheat" by huddling between grid cells to average out positive and negative heights rather than crossing discrete swap barriers. Switching to the discrete ME-ILS hierarchy—which combines 2-swaps, whole-row/column glides, Kempe-like $(2,2)$ plaquette loops, and threshold-leashed basin restarts—achieves 100.0% synthesis across every order from $n = 3$ to $n = 10$ (Table 8, up to $100! \approx 9.33 \times 10^{157}$ states) and explains why **5×5 is the hardest frustration bottleneck (8.03 restarts)** while larger grids ($n = 6 \ldots 10$, dropping to 3.34 restarts at $n = 10$) become easier to untangle as $f_{\text{null}}(n) = 1 - 2/(n-1)^2$ of the $(2,2)$ plaquette moves become diagonal-free (Figure 7).

2 Why Balancing a Plate Is Not Enough: The Zero-Mean Spectrum and the Impostor Square S_{low}

Suppose we place n^2 weights $Z_{ij} = 2s_{ij} - (n^2+1)$ on a rigid horizontal plate at the grid coordinates (X_j, Y_i) of Equation 1, as visualized in the *Energy Decomposition Explorer* (`energy-decomposition.html`). In Papers II and III [19, 18], we showed that if S is a magic square, then its 3D gravitational tilt moments $\langle X, Z\rangle = \sum_{i,j} X_j Z_{ij}$ and $\langle Y, Z\rangle = \sum_{i,j} Y_i Z_{ij}$ vanish identically, so the plate balances horizontally on its center of mass while its two main diagonals also balance ($R_{D_+} = R_{D_-} = 0$). When users first experiment with `physicalSolver`, a natural intuition is to ask whether the reverse is true: if we rearrange the numbers $1 \ldots n^2$ until the plate has zero X-tilt, zero Y-tilt, and exact magic sums along both main diagonals, does that force the square to be magic?

To see why the answer is a dramatic "no," we first compute the exact eigenvalue spectrum of the line-incidence operator $L_n = A_n^\top A_n$ both on the full Euclidean space $\mathbb{R}^{n^2}$ and on the (n^2-1)-dimensional zero-mean hyperplane $\mathbf{1}^\perp = \{\mathbf{z} \in \mathbb{R}^{n^2} : \mathbf{1}^\top \mathbf{z} = 0\}$ inhabited by every centred permutation vector $\mathbf{z}(S)$.

2.1 The Rook's-Graph Decomposition and Integer Spectrum on $\mathbf{1}^\perp$

Let $\mathbf{r}_1, \ldots, \mathbf{r}_n \in \{0,1\}^{n^2}$ denote the row-indicator vectors of the $n \times n$ grid, $\mathbf{c}_1, \ldots, \mathbf{c}_n \in \{0,1\}^{n^2}$ the column-indicator vectors, and $\mathbf{d}_+, \mathbf{d}_- \in \{0,1\}^{n^2}$ the main and anti-diagonal indicator vectors. Then the operator L_n splits into a row-and-column part L_{rc} plus a rank-2 diagonal part D:

$$L_n = A_n^\top A_n = L_{\text{rc}} + D, \qquad L_{\text{rc}} = \sum_{i=1}^{n} \mathbf{r}_i \mathbf{r}_i^\top + \sum_{j=1}^{n} \mathbf{c}_j \mathbf{c}_j^\top, \qquad D = \mathbf{d}_+ \mathbf{d}_+^\top + \mathbf{d}_- \mathbf{d}_-^\top. \tag{3}$$

In graph theory, $L_{\text{rc}} = A_{\text{Rook}} + 2I_{n^2}$ is the signless incidence operator of the $n \times n$ Rook's graph $K_n \square K_n = (K_n \otimes I_n) \oplus (I_n \otimes K_n)$, whose edges connect cells in the same row or column [12, 4]. Because the n rows partition the grid and the n columns also partition the grid, $\sum_{i=1}^{n} \mathbf{r}_i = \sum_{j=1}^{n} \mathbf{c}_j = \mathbf{1}_{n^2}$, reducing the rank of L_{rc} from $2n$ to $2n-1$. Adding the two independent main diagonals gives $\operatorname{rank}(A_n) = 2n+1$, leaving a magic nullspace $\ker(L_n) = \ker(A_n)$ of dimension

$\dim\ker(L_n) = n^2 - 2n - 1$, a classic dimension formula in the linear algebra of magic squares [31, 15, 17].

If we compute the eigenvalues of L_n numerically on $\mathbb{R}^{n^2}$, two ugly irrational numbers $\lambda_\pm(n)$ appear (Figure 1a, orange diamonds), because the constant vector $\mathbf{1}$ has non-zero overlap $\mathbf{1}^\top \mathbf{d}_\pm = n$ with the two diagonals. However, every centred permutation vector $\mathbf{z}(S)$ has zero sum ($\mathbf{1}^\top \mathbf{z} = 0$) and therefore lives entirely inside the zero-mean hyperplane $\mathbf{1}^\perp$. Restricting L_n to $\mathbf{1}^\perp$ decouples the constant mode and confines the two diagonals entirely inside the $(n-1)^2$-dimensional kernel of the Rook's graph, causing all irrational eigenvalues to collapse into clean integers (Figure 1a and Table 2):

Proposition 2.1 (Zero-Mean Integer Spectrum and Random-Shuffle Expectation of L_n)**.** *For every order $n \geq 3$, the line-incidence matrix $A_n \in \{0,1\}^{(2n+2)\times n^2}$ has rank $2n+1$, and its magic nullspace $\ker(L_n) = \ker(A_n) \subset \mathbf{1}^\perp$ has dimension*

$$\dim\ker(L_n) = n^2 - 2n - 1 = (n-1)^2 - 2. \tag{4}$$

On the ambient space $\mathbb{R}^{n^2}$, the $2n+1$ non-zero eigenvalues of L_n consist of two irrational roots $\lambda_\pm(n)$ together with integer eigenvalues:

$$\operatorname{spec}(L_n) = \begin{cases} \left\{0^{[n^2-2n-1]},\ \lambda_-(n)^{[1]},\ n^{[2n-1]},\ \lambda_+(n)^{[1]}\right\}, & \text{even } n \geq 4, \\ \left\{0^{[n^2-2n-1]},\ \lambda_-(n)^{[1]},\ (n-1)^{[1]},\ n^{[2n-2]},\ \lambda_+(n)^{[1]}\right\}, & \text{odd } n \geq 3, \end{cases} \tag{5}$$

where $\lambda_\pm(n) = \frac{1}{2}\big(3n \pm \sqrt{n(n+16)}\big)$ for even n and $\lambda_\pm(n) = \frac{1}{2}\big(3n+1 \pm \sqrt{n^2+14n+1}\big)$ for odd n. Let $P_{\mathbf{1}^\perp} = I_{n^2} - \frac{1}{n^2}\mathbf{1}\mathbf{1}^\top$ denote the orthogonal projector onto $\mathbf{1}^\perp$, and let $\widetilde{L}_n = P_{\mathbf{1}^\perp} L_n P_{\mathbf{1}^\perp}\big|_{\mathbf{1}^\perp}$ be the restricted operator on $\mathbf{1}^\perp$. On $\mathbf{1}^\perp$, the irrational roots disappear and all eigenvalues are integers:

$$\operatorname{spec}\big(\widetilde{L}_n\big|_{\mathbf{1}^\perp}\big) = \begin{cases} \left\{0^{[n^2-2n-1]},\ (n-2)^{[1]},\ n^{[2n-1]}\right\}, & \text{if } n \geq 4 \text{ is even}, \\ \left\{0^{[n^2-2n-1]},\ (n-1)^{[2]},\ n^{[2n-2]}\right\}, & \text{if } n \geq 3 \text{ is odd}. \end{cases} \tag{6}$$

Moreover, for every integer $n \geq 3$ (both odd and even), $\operatorname{tr}(\widetilde{L}_n|_{\mathbf{1}^\perp}) = 2(n^2-1)$, so the average eigenvalue of $\widetilde{L}_n$ on $\mathbf{1}^\perp$ is $\bar{\mu}(\widetilde{L}_n) = \operatorname{tr}(\widetilde{L}_n|_{\mathbf{1}^\perp})/(n^2-1) = 2$, and the expected line-indicator energy of a uniform random shuffle $S \sim \mathrm{Unif}(S_{n^2})$ with $\|\mathbf{z}\|^2 = \frac{1}{3}n^2(n^4-1)$ is exactly twice the height variance sum:

$$\mathbb{E}_{S\sim\mathrm{Unif}(S_{n^2})}\big[\mathcal{H}_{\mathrm{line}}(S)\big] = 2\,\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1). \tag{7}$$

Proof. By the standard spectral decomposition of the Rook's graph $K_n \square K_n$ [12, 4], L_{rc} splits $\mathbb{R}^{n^2}$ into three mutually orthogonal eigenspaces: the constant line $\operatorname{span}\{\mathbf{1}\}$ with eigenvalue $2n$ (multiplicity 1), the $(2n-2)$-dimensional zero-mean row and column space $\mathcal{V}_{\mathrm{row}} \oplus \mathcal{V}_{\mathrm{col}}$ with flat eigenvalue n (where $\mathcal{V}_{\mathrm{row}} = \{\mathbf{u} \otimes \mathbf{1}_n : \mathbf{u}^\top \mathbf{1}_n = 0\}$ and $\mathcal{V}_{\mathrm{col}} = \{\mathbf{1}_n \otimes \mathbf{v} : \mathbf{v}^\top \mathbf{1}_n = 0\}$), and the $(n-1)^2$-dimensional tensor-product kernel $\ker(L_{\mathrm{rc}}) = \mathcal{V}_{\mathrm{row}} \otimes \mathcal{V}_{\mathrm{col}}$ with eigenvalue 0.

Because every row and column crosses each main diagonal in exactly one cell, $\mathbf{r}_i^\top \mathbf{d}_\pm = \mathbf{c}_j^\top \mathbf{d}_\pm = 1$ for all i, j. Thus the zero-mean projections $P_{\mathbf{1}^\perp}\mathbf{d}_\pm = \mathbf{d}_\pm - \frac{1}{n}\mathbf{1}$ have zero row and column sums and lie entirely inside $\ker(L_{\mathrm{rc}}) = \mathcal{V}_{\mathrm{row}} \otimes \mathcal{V}_{\mathrm{col}}$. Consequently, $\mathcal{V}_{\mathrm{row}} \oplus \mathcal{V}_{\mathrm{col}} \subset \ker(D)$ remains an exact eigenspace of both L_n and $\widetilde{L}_n$ with eigenvalue n and multiplicity $2n-2$.

On the remaining 3-dimensional subspace $\operatorname{span}\{\mathbf{1}, \mathbf{d}_+, \mathbf{d}_-\}$, define the symmetric and antisymmetric diagonal vectors $\mathbf{s}_d = \mathbf{d}_+ + \mathbf{d}_-$ and $\mathbf{a}_d = \mathbf{d}_+ - \mathbf{d}_-$. Since $\mathbf{a}_d \in \ker(L_{\mathrm{rc}})$ and $\|\mathbf{a}_d\|^2 = 2n - 2\mathbf{d}_+^\top \mathbf{d}_-$, we have $D\mathbf{a}_d = n\mathbf{a}_d$ for even n (where the two diagonals are disjoint, $\mathbf{d}_+^\top \mathbf{d}_- = 0$) and $D\mathbf{a}_d = (n-1)\mathbf{a}_d$ for odd n (where the two diagonals share the center cell, $\mathbf{d}_+^\top \mathbf{d}_- = 1$). On $\operatorname{span}\{\mathbf{1}, \mathbf{s}_d\}$, L_n has the 2×2 matrix $[L_n]_{\mathcal{B}} = \left[\begin{smallmatrix} 2n & 4 \\ n & n \end{smallmatrix}\right]$ for even n and $\left[\begin{smallmatrix} 2n & 4 \\ n & n+1 \end{smallmatrix}\right]$ for odd n, whose characteristic polynomials give the irrational roots $\lambda_\pm(n)$ in Equation 5. Projecting onto $\mathbf{1}^\perp$ replaces $\mathbf{s}_d$ by $\widetilde{\mathbf{s}}_d = \mathbf{s}_d - \frac{2}{n}\mathbf{1} \in \ker(L_{\mathrm{rc}})$, which has squared norm $\|\widetilde{\mathbf{s}}_d\|^2 = \|\mathbf{s}_d\|^2 - 4 = 2(n-2)$

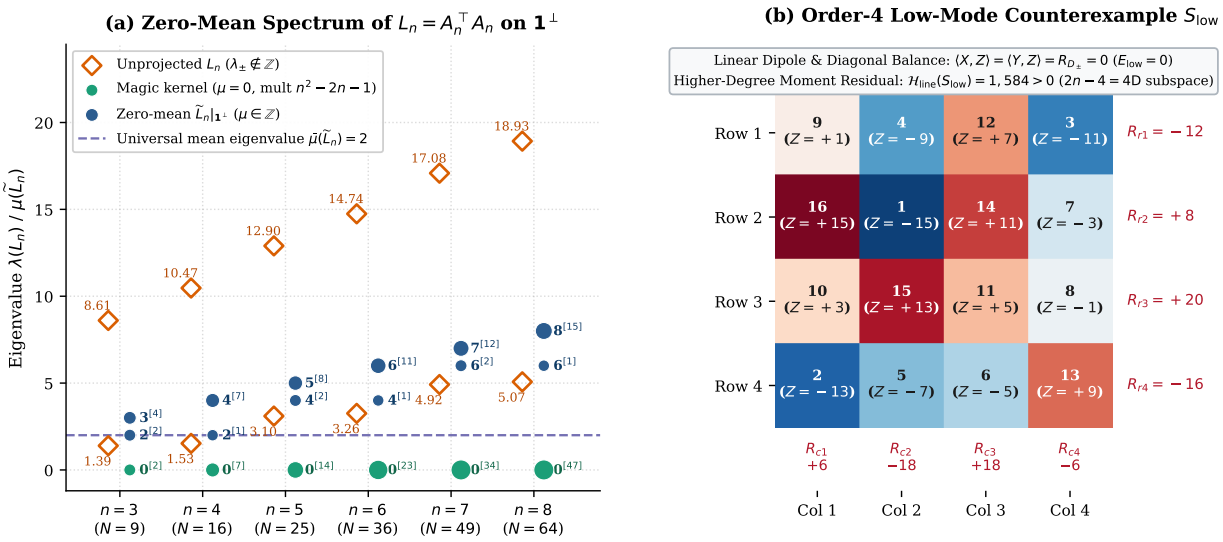


Figure 1: **Zero-mean integer spectrum of the line-incidence operator $L_n = A_n^\top A_n$ and the 4×4 "impostor square" S_{low}. (a)** Comparison of the unprojected eigenvalues $\lambda(L_n)$ on $\mathbb{R}^{n^2}$ (hollow orange diamonds, showing the irrational branches $\lambda_\pm(n)$) against the zero-mean restricted spectrum $\mu(\widetilde{L}_n|_{\mathbf{1}^\perp})$ (filled navy and green circles, labeled $\mu^{[\text{mult}]}$) for orders $n = 3 \ldots 8$. Restricting to $\mathbf{1}^\perp$ yields exact integers with universal mean $\bar{\mu}(\widetilde{L}_n) = 2$ (dashed purple line). **(b)** Explicit 4×4 counterexample S_{low} (Remark 2.2) from `energy-decomposition.html` whose 3D tilt moments $\langle X, Z\rangle = \langle Y, Z\rangle = 0$ and main diagonals $R_{D_\pm} = 0$ vanish ($E_{\text{low}} = 0$), yet whose alternating row and column sums produce $\mathcal{H}_{\text{line}}(S_{\text{low}}) = 1{,}584 > 0$.

Table 2: Spectral invariants of the unprojected operator L_n on $\mathbb{R}^{n^2}$ and zero-mean operator $\widetilde{L}_n$ on $\mathbf{1}^\perp$ across orders $n = 3 \ldots 8$, verified by `verify_paper_d.py`.

n	$N = n^2$	$\operatorname{rank}(A_n)$	$\dim\ker(L_n)$	Irrational $\lambda_\pm(L_n)$	Zero-Mean $\operatorname{spec}(\widetilde{L}_n|_{\mathbf{1}^\perp})$	$\operatorname{tr}(\widetilde{L}_n)$	$\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2$
3	9	7	2	1.3944, 8.6056	$\{0^{[2]}, 2^{[2]}, 3^{[4]}\}$	16	480
4	16	9	7	1.5279, 10.4721	$\{0^{[7]}, 2^{[1]}, 4^{[7]}\}$	30	2,720
5	25	11	14	3.1010, 12.8990	$\{0^{[14]}, 4^{[2]}, 5^{[8]}\}$	48	10,400
6	36	13	23	3.2554, 14.7446	$\{0^{[23]}, 4^{[1]}, 6^{[11]}\}$	70	31,080
7	49	15	34	4.9172, 17.0828	$\{0^{[34]}, 6^{[2]}, 7^{[12]}\}$	96	78,400
8	64	17	47	5.0718, 18.9282	$\{0^{[47]}, 6^{[1]}, 8^{[15]}\}$	126	174,720

for even n and $2(n-1)$ for odd n. Because $\widetilde{\mathbf{s}}_d$ and $\mathbf{a}_d$ are orthogonal vectors inside $\ker(L_{\text{rc}})$, restricting $D = \frac{1}{2}(\mathbf{s}_d\mathbf{s}_d^\top + \mathbf{a}_d\mathbf{a}_d^\top)$ to $\mathbf{1}^\perp$ yields integer eigenvalues $\frac{1}{2}\|\widetilde{\mathbf{s}}_d\|^2 = n - 2$ and $\frac{1}{2}\|\mathbf{a}_d\|^2 = n$ for even n, or $n-1$ and $n-1$ for odd n, while their orthogonal complement in $\mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$ has dimension $(n-1)^2 - 2 = n^2 - 2n - 1$, proving Equation 4–Equation 6.

Finally, Equation 7 can be seen in two complementary ways. From the trace, $\operatorname{tr}(L_n) = (2n+2)n = 2n(n+1)$ and $\hat{\mathbf{1}}^\top L_n \hat{\mathbf{1}} = 2n+2$, so $\operatorname{tr}(\widetilde{L}_n|_{\mathbf{1}^\perp}) = 2n(n+1) - 2(n+1) = 2(n^2-1)$ and $\bar{\mu}(\widetilde{L}_n) = 2$. From elementary probability, any line $L \in \mathcal{L}_n$ of length n in a random shuffle is a simple random sample of n values drawn without replacement from a zero-sum population of n^2 values, whose sum has variance $\mathbb{E}[R_L(S)^2] = n\left(\frac{n^2-n}{n^2-1}\right)\frac{\|\mathbf{z}\|^2}{n^2} = \frac{\|\mathbf{z}\|^2}{n+1}$ [9]. Multiplying by the $2n+2 = 2(n+1)$ lines cancels the $n+1$ in the denominator and gives $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$. □

2.2 The 4×4 Impostor Square S_{low} and the $2n-4$ Hidden Polynomial Modes

With Proposition 2.1 in hand, we can immediately see why balancing a plate's 3D tilt and diagonals is not enough to find a magic square:

Remark 2.2 (The Balanced Impostor Square S_{low} and the $2n-4$ Hidden Modes). In the *Energy Decomposition Explorer* (`energy-decomposition.html`), the 4-mode tilt-plus-diagonal energy is

defined as

$$E_{\text{low}}(S) = \langle X, Z\rangle^2 + \langle Y, Z\rangle^2 + R_{D_+}(S)^2 + R_{D_-}(S)^2. \tag{8}$$

Because the linear coordinate vectors $X \in \mathcal{V}_{\text{col}}$ and $Y \in \mathcal{V}_{\text{row}}$ are degree-1 column and row moments, they are mutually orthogonal and orthogonal to the projected diagonals $\widetilde{\mathbf{d}}_\pm \in \mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$. Thus $\{X, Y, \widetilde{\mathbf{d}}_+, \widetilde{\mathbf{d}}_-\}$ spans only a 4-dimensional slice of the $2n$-dimensional active image $\text{im}(\widetilde{L}_n) = \mathcal{V}_{\text{row}} \oplus \mathcal{V}_{\text{col}} \oplus \text{span}\{\widetilde{\mathbf{s}}_d, \mathbf{a}_d\} \subset \mathbf{1}^\perp$. Its orthogonal complement inside $\text{im}(\widetilde{L}_n)$ has dimension

$$\dim\big(\text{im}(\widetilde{L}_n) \cap \ker(E_{\text{low}})\big) = 2n - 4 = 2(n-2), \tag{9}$$

spanned by the $n-2$ higher-degree column polynomial moments $P_k(X) \in \mathcal{V}_{\text{col}}$ and $n-2$ row polynomial moments $P_k(Y) \in \mathcal{V}_{\text{row}}$ of degrees $2 \le k \le n-1$, all sharing eigenvalue $\mu = n$. At order $n = 4$ ($2n - 4 = 4$ hidden quadratic and cubic modes $\langle X^2, Z\rangle, \langle X^3, Z\rangle, \langle Y^2, Z\rangle, \langle Y^3, Z\rangle$), loading the preset square S_{low} in `energy-decomposition.html` (Figure 1b),

$$S_{\text{low}} = \begin{bmatrix} 9 & 4 & 12 & 3 \\ 16 & 1 & 14 & 7 \\ 10 & 15 & 11 & 8 \\ 2 & 5 & 6 & 13 \end{bmatrix} \in S_{16}, \tag{10}$$

shows a plate that balances with zero X-tilt ($\langle X, Z\rangle = -3(6) - 1(-18) + 1(18) + 3(-6) = 0$), zero Y-tilt ($\langle Y, Z\rangle = 3(-12)+1(8)-1(20)-3(-16) = 0$), and exact magic diagonals ($9+1+11+13 = 34$ and $3+14+15+2 = 34$, so $E_{\text{low}}(S_{\text{low}}) = 0$). Yet its rows sum to $(28, 38, 44, 26)$ and its columns sum to $(37, 25, 43, 31)$, hiding a massive line-sum energy $\mathcal{H}_{\text{line}}(S_{\text{low}}) = (-12)^2 + 8^2 + 20^2 + (-16)^2 + 6^2 + (-18)^2 + 18^2 + (-6)^2 = 1{,}584 > 0$ entirely inside the quadratic and cubic row/column alternations!

Proposition 2.1 also defines a geometric ratio that will prove decisive when we compare different grid sizes n in Section 5, namely the *nullspace-to-constraint ratio* on $\mathbf{1}^\perp$:

$$\eta_{\text{MC}}(n) = \frac{\dim\ker(\widetilde{L}_n)}{\text{rank}(\widetilde{L}_n)} = \frac{n^2 - 2n - 1}{2n}, \tag{11}$$

analogous to the ratio of internal zero-energy "floppy" degrees of freedom to active linear constraints in Maxwell–Calladine frame counting [22, 5]. In continuous linear algebra on $\mathbf{1}^\perp$, $\dim\ker(\widetilde{L}_n) = n^2 - 2n - 1$ is already positive $(2, 7, 14, \dots)$ for all $n \ge 3$. What makes small magic squares ($n = 3, 4, 5$) hard to find by random swapping is the *discrete permutation constraint* $\mathbf{z} \in S_{n^2}$: we must land on a discrete corner of the permutahedron that happens to lie inside $\ker(\widetilde{L}_n)$ when the nullspace dimension is smaller than or comparable to the number of constraints ($\eta_{\text{MC}}(3) = 0.33$, $\eta_{\text{MC}}(4) = 0.88$, $\eta_{\text{MC}}(5) = 1.40$). As $n \ge 6$ grows, the magic nullspace expands quadratically ($\dim\ker = 79$ at $n = 10$) while the constraint rank grows only linearly ($2n = 20$ at $n = 10$, $\eta_{\text{MC}}(10) = 3.95$), giving discrete multi-exchange search room to maneuver without breaking the diagonals.

3 Inside the Magic Manifold: Window Entropies, Panmagic Quartets, and the 56 Octupole Zeros

Once a permutation reaches the magic ground-state manifold $\ker(\widetilde{L}_n)$ where $\mathcal{H}_{\text{line}}(S) = 0$, all $2n + 2$ global 1D line sums are identical. Yet anyone who cycles through the 880 Frénicle order-4 magic squares in the interactive *Energy Decomposition Explorer* (`energy-decomposition.html`) notices immediately that different magic squares look and behave very differently as 2D textures and 3D Magic Gems [10, 24, 6]. Some squares distribute high and low numbers in a crisp, uniform checkerboard across every 2×2 neighborhood, whereas others cluster extreme heights into corner blocks or exhibit a pronounced three-dimensional octupole twist. To quantify what the eye perceives in the browser explorer, we evaluate sliding $k \times k$ sub-window energies and Shannon entropies exhaustively across all 880 Frénicle order-4 magic squares and higher-order constructive families.

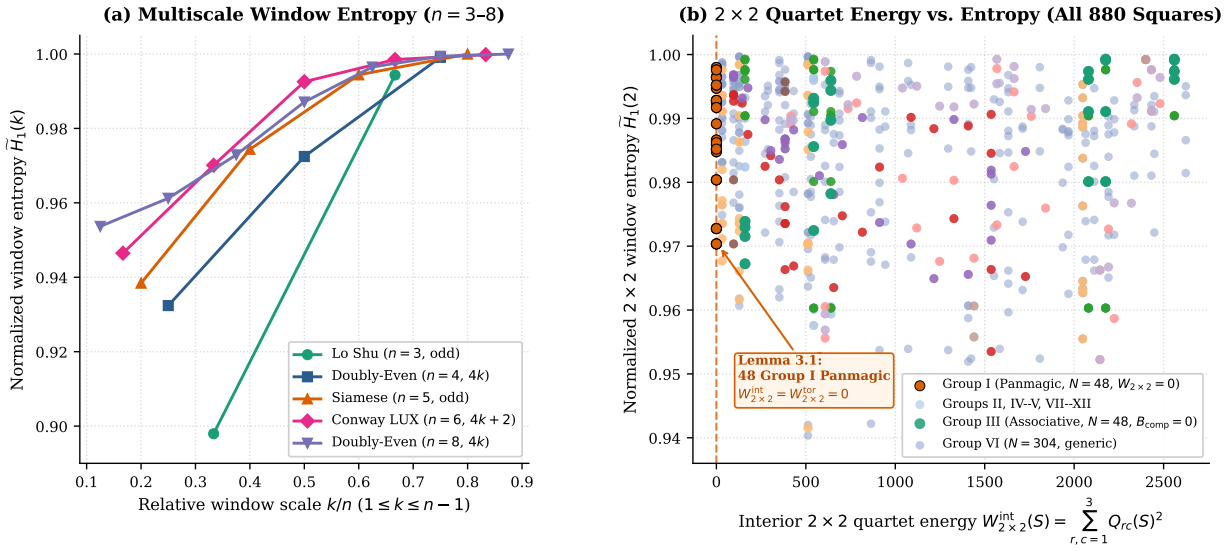


Figure 2: **Multiscale sliding-window Shannon entropy and 2×2 quartet energy census. (a)** Normalized ℓ_1 window entropy $\widetilde{H}_1(k)$ versus relative scale k/n across five constructive magic-square families ($n = 3$ Lo Shu, $n = 4, 8$ Doubly-Even, $n = 5$ Siamese, and $n = 6$ Conway LUX). **(b)** Joint distribution of interior 2×2 quartet residual energy $W^{\text{int}}_{2\times 2}(S)$ and window entropy $\widetilde{H}_1(2)$ across all 880 Frénicle order-4 magic squares partitioned by Dudeney Groups I–XII, illustrating Lemma 3.1: $W^{\text{int}}_{2\times 2}(S) = 0$ holds if and only if $W^{\text{tor}}_{2\times 2}(S) = 0$, isolating the 48 panmagic squares of Group I.

3.1 Sliding-Window Shannon Entropy and Constructive Family Scaling

For any window width $k \in \{1, \ldots, n\}$, an $n \times n$ grid contains $M_k = (n-k+1)^2$ interior contiguous $k \times k$ sub-windows $W^{(k)}_{r,c} = \{r, \ldots, r+k-1\} \times \{c, \ldots, c+k-1\}$ $(1 \le r, c \le n-k+1)$. In `energy-decomposition.html`, the user can toggle the window size k and norm exponent $p \in \{1, 2, 4\}$ to inspect how evenly the centred heights Z_{ij} are spread across local patches. Formally, the local ℓ^p_p window weight and normalized Shannon window entropy are defined by

$$w_p\big(W^{(k)}_{r,c}\big) = \sum_{(i,j)\in W^{(k)}_{r,c}} |Z_{ij}|^p, \qquad p^{(k)}_{r,c} = \frac{w_p(W^{(k)}_{r,c})}{\sum_{r',c'} w_p(W^{(k)}_{r',c'})}, \qquad \widetilde{H}_p(k) = \frac{-\sum_{r,c} p^{(k)}_{r,c} \log_2 p^{(k)}_{r,c}}{\log_2\big((n-k+1)^2\big)}. \quad (12)$$

Here $\widetilde{H}_p(k) \in (0, 1]$ measures the spatial uniformity of height magnitude across all $k \times k$ sub-windows, reaching $\widetilde{H}_p(k) = 1$ if and only if every $k \times k$ window carries the exact same ℓ^p_p weight.

Figure 2a and Table 3 compare the multiscale entropy profile $\widetilde{H}_p(k)$ across five classical constructive presets included in the interactive explorer: odd-order de La Loubère (Siamese) squares ($n = 3, 5$) [8], doubly-even 4×4 diagonal-mask squares ($n = 4, 8$) [1], and Conway's LUX singly-even construction ($n = 6$) [2]. At $k = 2$, the 3×3 Lo Shu square achieves the highest local uniformity ($\widetilde{H}_1(2) = 0.99435$, $\widetilde{H}_2(2) = 0.99073$), whereas doubly-even mask constructions at $n = 4, 8$ concentrate extreme values symmetrically in corner and central blocks, lowering $\widetilde{H}_2(2)$ to 0.91498 ($n = 4$) and 0.90598 ($n = 8$). While the $n = 3, 4, 5, 8$ constructions are centrosymmetric ($B_{\text{comp}} = 0$ complement breaks and $M_{xyz} = 0$), Conway's $n = 6$ LUX method breaks all 18/18 antipodal complement pairs and generates a non-zero mixed octupole moment $M_{xyz} = -132$.

3.2 Why Interior 2×2 Balance Forces Toroidal Panmagic Balance

Beyond magnitude entropy, the signed sum of each 2×2 quartet $Q_{r,c}(S) = Z_{r,c} + Z_{r+1,c} + Z_{r,c+1} + Z_{r+1,c+1}$ (with indices taken modulo n in the toroidal case) defines the *interior* and *toroidal* 2×2

Table 3: Multiscale window entropy $\widetilde{H}_p(2)$, interior and toroidal 2×2 quartet energies $W_{2\times 2}$, complement breaks B_{comp}, and mixed octupole moment M_{xyz} across constructive magic square families ($n = 3\ldots 8$).

Family	Order / Class	$\widetilde{H}_1(2)$	$\widetilde{H}_2(2)$	$W^{\text{int}}_{2\times 2}(S)$	$W^{\text{tor}}_{2\times 2}(S)$	Breaks B_{comp}	Octupole M_{xyz}
Lo Shu (Siamese)	$n = 3$ (odd)	0.99435	0.99073	80	240	0/4	0
Doubly-Even Mask	$n = 4$ ($4k$)	0.97245	0.91498	544	1,088	0/8	0
Siamese (de La Loubère)	$n = 5$ (odd)	0.97438	0.94254	16,320	25,200	0/12	0
Conway LUX Method	$n = 6$ ($4k+2$)	0.97011	0.93161	84,640	97,440	18/18	−132
Doubly-Even Mask	$n = 8$ ($4k$)	0.96123	0.90598	112,320	149,760	0/32	0

quartet energies displayed in `energy-decomposition.html`:

$$W^{\text{int}}_{2\times 2}(S) = \sum_{r=1}^{n-1}\sum_{c=1}^{n-1} Q_{r,c}(S)^2, \qquad W^{\text{tor}}_{2\times 2}(S) = \sum_{r=1}^{n}\sum_{c=1}^{n} Q_{r,c}(S)^2. \tag{13}$$

When exploring even-order magic squares in the browser, one observes a striking phenomenon: whenever the $(n-1)^2$ interior 2×2 windows are balanced ($W^{\text{int}}_{2\times 2}(S) = 0$), the $2n-1$ boundary-wrapping windows across the edges and corners of the torus automatically balance as well ($W^{\text{tor}}_{2\times 2}(S) = 0$). An elementary row- and column-pair partition explains why:

Lemma 3.1 (Interior–Toroidal 2×2 Quartet Equivalence)**.** *Let S be any magic square of even order $n = 2m \geq 4$. Then its $(2m-1)^2$ interior 2×2 blocks have zero centred sum ($W^{\text{int}}_{2\times 2}(S) = 0$)* if and only if *all $4m^2$ toroidal 2×2 blocks have zero centred sum ($W^{\text{tor}}_{2\times 2}(S) = 0$). In particular, across the* 880 *Frénicle magic squares of order $n = 4$, vanishing of the* 9 *interior quartets ($W^{\text{int}}_{2\times 2}(S) = 0$) is equivalent to vanishing of all* 16 *toroidal quartets ($W^{\text{tor}}_{2\times 2}(S) = 0$) and holds on* exactly the 48 panmagic (diabolic) squares of Dudeney Group I *(Figure 2b and Table 4).*

Proof. Clearly $W^{\text{tor}}_{2\times 2}(S) = 0 \implies W^{\text{int}}_{2\times 2}(S) = 0$. Conversely, suppose S is a magic square of even order $n = 2m$ ($\mathcal{H}_{\text{line}}(S) = 0$) with $Q_{r,c}(S) = 0$ for all interior blocks $1 \leq r, c \leq 2m-1$. For any horizontally wrapping 2×2 block $Q_{r,2m}(S)$ ($1 \leq r \leq 2m-1$), the sum of rows r and $r+1$ vanishes by magic row balance and partitions into $m-1$ interior 2×2 blocks starting at even columns plus $Q_{r,2m}(S)$:

$$0 = R_{r_r}(S) + R_{r_{r+1}}(S) = \sum_{j=1}^{m-1} Q_{r,2j}(S) + Q_{r,2m}(S). \tag{14}$$

Since $Q_{r,2j}(S) = 0$ for $1 \leq j \leq m-1$, Equation 14 forces $Q_{r,2m}(S) = 0$. By the same argument on columns c and $c+1$, $Q_{2m,c}(S) = 0$ for $1 \leq c \leq 2m-1$, and finally $Q_{2m,2m}(S) = (R_{r_1}(S) + R_{r_{2m}}(S)) - \sum_{j=1}^{m-1} Q_{2m,2j}(S) = 0$. At $n = 4$ ($m = 2$), $W^{\text{tor}}_{2\times 2}(S) = 0$ is a classical characterization of the 48 panmagic squares of Dudeney Group I [10, 24]. □

3.3 Complement Parity and the Puzzle of the 56 Octupole Zeros

In Paper II [19] and Paper III [18], the mixed third-order moment (octupole) $M_{xyz}(S) = \sum_{i,j=1}^{n} X_j Y_i Z_{ij}$ governed the three-dimensional chiral twist of a Magic Gem and the asymmetry between its static and dynamic axes. To connect $M_{xyz}(S)$ to discrete cell pairing, define the *complement-break count* $B_{\text{comp}}(S)$ as the number of antipodal cell pairs $\{(i,j),(n+1-i,n+1-j)\}$ that fail the associative condition $Z_{ij} + Z_{n+1-i,n+1-j} = 0$ ($s_{ij} + s_{n+1-i,n+1-j} = n^2+1$). When we scanned $M_{xyz}(S)$ across all 880 order-4 squares in `energy-decomposition.html`, we expected $M_{xyz}(S) = 0$ to occur on the 48 centrosymmetric (associative) squares of Dudeney Group III. Surprisingly, the census returned 56 zeros instead of 48: 8 non-associative squares in Groups IX and X also had $M_{xyz}(S) = 0$! Decomposing the 4×4 grid into concentric shells resolves this puzzle completely:

Proposition 3.2 (Concentric Shell Quadrupole Identity and Octupole Zero Classification)**.** *For any order-4 magic square S, partition the* 16 *cells into the* 4 *outer corners ($|X_j Y_i| = 9$), the* 8

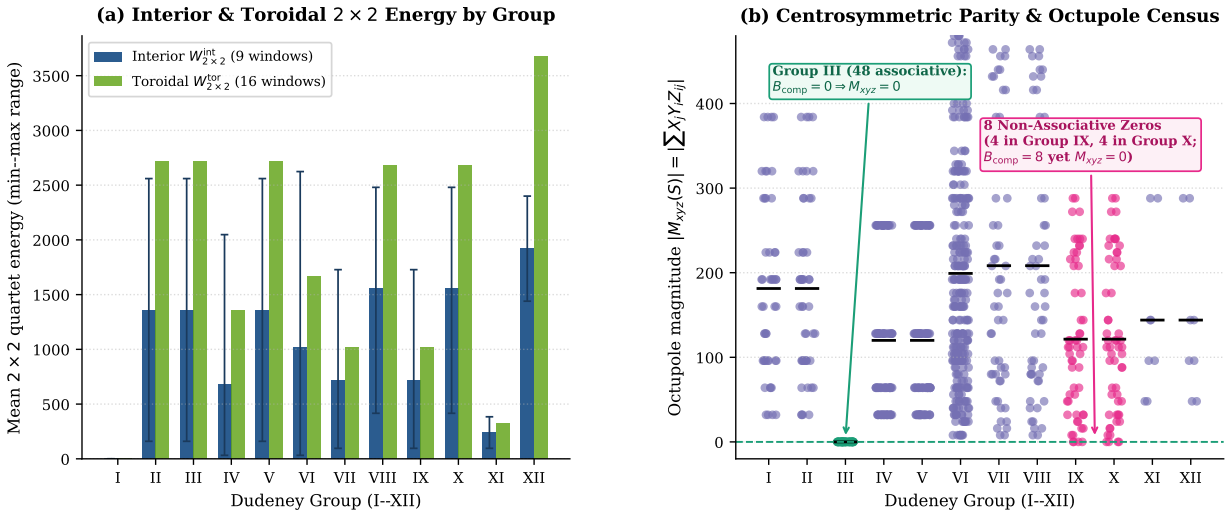


Figure 3: **Group-resolved 2×2 quartet energies and mixed octupole census across Dudeney Groups I–XII. (a)** Mean interior $W^{\text{int}}_{2\times2}$ (9 windows, navy bars with min–max range) and toroidal $W^{\text{tor}}_{2\times2}$ (16 windows, green bars) across the 12 Dudeney groups. **(b)** Magnitude of the mixed octupole moment $|M_{xyz}(S)| = |\sum X_j Y_i Z_{ij}|$ across all 880 squares (Proposition 3.2), highlighting the 48 associative squares of Group III ($B_{\text{comp}} = 0 \implies M_{xyz} = 0$) and the 8 anomalous non-associative octupole-zero squares in Groups IX and X ($B_{\text{comp}} = 8$ yet $M_{xyz} = 0$ via the Concentric Shell Quadrupole resonance $8Q_{\text{corner}} = -3Q_{\text{edge}}$).

non-corner boundary edge cells ($|X_jY_i| = 3$), and the 4 inner 2×2 cells ($|X_jY_i| = 1$), with signed quadrupolar moments

$$Q_{\text{corner}}(S) = (Z_{11} + Z_{44}) - (Z_{14} + Z_{41}), \qquad Q_{\text{inner}}(S) = (Z_{22} + Z_{33}) - (Z_{23} + Z_{32}),$$
$$Q_{\text{edge}}(S) = \big[(Z_{12} + Z_{43}) - (Z_{13} + Z_{42})\big] + \big[(Z_{21} + Z_{34}) - (Z_{31} + Z_{24})\big]. \tag{15}$$

Across all 880 *Frénicle magic squares of order* 4*, the complement-break count* $B_{\text{comp}}(S) \in \{0, 1, \ldots, 8\}$ *takes* only the two extreme values $\{0, 8\}$*:* $B_{\text{comp}}(S) = 0$ *on the* 48 *associative squares of Dudeney Group III, and* $B_{\text{comp}}(S) = 8$ *on all remaining* 832 *squares, where the* 8 *complement pairs follow non-antipodal regular* 2*-factors of the* 4×4 *grid [10]. Moreover, diagonal balance* $R_{D_+}(S) = R_{D_-}(S) = 0$ *forces the exact concentric identity* $Q_{\text{inner}}(S) = -Q_{\text{corner}}(S)$*, reducing the mixed octupole moment of every* 4×4 *magic square to the Diophantine linear form*

$$M_{xyz}(S) = -9\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S) - Q_{\text{inner}}(S) = -8\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S). \tag{16}$$

Consequently, $M_{xyz}(S) = 0$ *holds if and only if* $8\,Q_{\text{corner}}(S) = -3\,Q_{\text{edge}}(S)$*, which is satisfied on* exactly 56 of the 880 squares *(Figure 3b and Table 4). First, on the* 48 *associative squares of Group III, centrosymmetric parity* $Z_{5-i,5-j} = -Z_{ij}$ *forces* $Q_{\text{corner}}(S) = Q_{\text{edge}}(S) = 0$ *identically, yielding* 48 *symmetry-protected zeros. Second, across the* 288 *squares of Groups I, II, IV, and V, the complement pairings along rows, columns, or* 2×2 *blocks force* $Q_{\text{edge}}(S) = 0$ *while* $Q_{\text{corner}}(S) \neq 0$*, so* $M_{xyz}(S) = -8\,Q_{\text{corner}}(S) \neq 0$ *can never vanish in those groups. Third, the remaining* 8 *octupole zeros arise from coprime shell resonances among non-associative squares ($B_{\text{comp}} = 8$): specifically,* 4 *squares in Group IX (Frénicle* `#200, #208, #390, #711`*) and* 4 *squares in Group X (*`#492, #494, #794, #811`*) satisfy* $\big(Q_{\text{corner}}(S), Q_{\text{edge}}(S)\big) = (-3k, 8k)$ *at* $k = \pm4$ *(*`#208, #711, #492, #494`*) and* $k = -8$ *(*`#200, #390, #794, #811`*), mapped bijectively between Groups IX and X by the simultaneous row/column parity permutation* $\pi = (2, 4, 1, 3)$ *and paired within each group by complementation* $S \leftrightarrow 171 - S$*.*

Proof. Since $R_{D_+}(S) = (Z_{11} + Z_{44}) + (Z_{22} + Z_{33}) = 0$ and $R_{D_-}(S) = (Z_{14} + Z_{41}) + (Z_{23} + Z_{32}) = 0$, subtracting the two equations yields $Q_{\text{inner}}(S) = (Z_{22} + Z_{33}) - (Z_{23} + Z_{32}) = -(Z_{11} + Z_{44}) + (Z_{14} + Z_{41}) = -Q_{\text{corner}}(S)$. Expanding $M_{xyz}(S) = \sum_{i,j=1}^{4} X_j Y_i Z_{ij}$ with $(X_1, X_2, X_3, X_4) = (-3, -1, 1, 3)$

Table 4: Complete multiscale window and multipole census across the 12 Dudeney groups of the 880 order-4 magic squares, verified by `verify_paper_d.py`.

Dudeney Group	N_g	B_{comp}	Mean $\widetilde{H}_1(2)$	Mean $\widetilde{H}_2(2)$	$W^{\text{int}}_{2\times2}$ (mean [min, max])	$W^{\text{tor}}_{2\times2}$ (mean)	$\lvert M_{xyz}\rvert$ (mean, # = 0)
I (Panmagic)	48	8	0.98722	0.96742	0.0 [0, 0]	0.0	181.3 (0/48)
II (Semi-pan.)	48	8	0.98148	0.95260	1360.0 [160, 2560]	2720.0	181.3 (0/48)
III (Assoc.)	48	0	0.98807	0.96789	1360.0 [160, 2560]	2720.0	0.0 (48/48)
IV	96	8	0.97919	0.94745	680.0 [32, 2048]	1360.0	120.0 (0/96)
V	96	8	0.98831	0.96951	1360.0 [160, 2560]	2720.0	120.0 (0/96)
VI	304	8	0.98356	0.95870	1017.7 [32, 2624]	1671.2	199.2 (0/304)
VII	56	8	0.97874	0.94983	720.6 [96, 1728]	1015.4	208.3 (0/56)
VIII	56	8	0.98127	0.95081	1554.9 [416, 2480]	2684.0	208.3 (0/56)
IX	56	8	0.98432	0.95886	720.6 [96, 1728]	1015.4	121.4 (4/56)
X	56	8	0.98705	0.96844	1554.9 [416, 2480]	2684.0	121.4 (4/56)
XI	8	8	0.98516	0.96109	240.0 [96, 384]	320.0	144.0 (0/8)
XII	8	8	0.97871	0.94441	1920.0 [1440, 2400]	3680.0	144.0 (0/8)

and $(Y_1, Y_2, Y_3, Y_4) = (3, 1, -1, -3)$ gives $M_{xyz}(S) = -9\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S) - Q_{\text{inner}}(S) = -8\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S)$. Because $\gcd(8, 3) = 1$, vanishing of $M_{xyz}(S)$ requires $Q_{\text{corner}}(S) = -3k$ and $Q_{\text{edge}}(S) = 8k$ for some integer $k \in \mathbb{Z}$, verified exhaustively across all 880 squares in `verify_paper_d.py`. □

Across all 880 squares, the global maximum of 2×2 window entropy is achieved by Frénicle square `#123` in Group VI ($\widetilde{H}_1(2) = 0.999669$), while the global minimum occurs at square `#599` in Group VI ($\widetilde{H}_1(2) = 0.940345$). Notice also in Table 4 and Figure 3a that Groups II, III, and V share the exact same mean interior quartet energy $\langle W^{\text{int}}_{2\times2}\rangle = 1{,}360 = \|\mathbf{z}\|^2$ and mean toroidal quartet energy $\langle W^{\text{tor}}_{2\times2}\rangle = 2{,}720 = 2\|\mathbf{z}\|^2$.

4 The 3×3 Combination Lock: Why Pair and Triplet Swaps Get Trapped

Whereas Section 2 showed that the continuous quadratic form $\mathbf{z}^\top L_n \mathbf{z}$ on $\mathbf{1}^\perp$ is a smooth, convex paraboloid with no barriers, restricting $\mathbf{z}$ to the discrete permutation grid S_{n^2} turns that bowl into a rugged combinatorial maze [7, 28]. In the interactive *Physical Magic Square Lab* (`physics-lab.html`) and *Vector Force Field Simulator* (`vector-physics.html`), users can watch individual tiles swap places as they try to relieve line-sum stress. Almost immediately, a puzzling behavior appears even on a 3×3 board: about one quarter of all random starting boards freeze in a configuration where every row, the middle column, and both diagonals sum to 15, while the left column sums to 14 and the right column sums to 16. Clicking any pair of tiles—or even rotating any triplet of tiles—only makes the energy worse. We now explain the exact algebra of this "combination lock" on S_9 ($n = 3$) and its proliferation on S_{16} ($n = 4$).

4.1 The Self-Energy Penalty of a Single Swap

Let $\mathbf{g}(S) = \frac{1}{2}\nabla\mathcal{H}_{\text{line}}(\mathbf{z}) = L_n\mathbf{z} = A_n^\top \mathbf{R}(S) \in \mathbb{Z}^{n^2}$ denote the *cell-stress vector* rendered as force arrows in `vector-physics.html`, whose component $g_a(S) = \sum_{L \ni a} R_L(S)$ at cell $a \in \{1, \ldots, n^2\}$ is the sum of the centred residuals of the 2, 3, or 4 magic lines passing through cell a. Swapping the values at cells a and b via the 2-exchange transposition $\tau_{ab} \in S_{n^2}$ replaces $\mathbf{z}$ by $\mathbf{z}' = \mathbf{z} - d(\mathbf{e}_a - \mathbf{e}_b)$, where $d = Z_a - Z_b = 2(s_a - s_b)$. Expanding $\mathbf{z}'^\top L_n \mathbf{z}'$ yields the exact rank-1 quadratic assignment difference [28]:

Lemma 4.1 (Quadratic Transposition Difference Formula)**.** *For any permutation $S \in S_{n^2}$ and any pair of cells $a \neq b$ with height difference $d = Z_a - Z_b$, the change in the line-indicator Hamiltonian under the transposition τ_{ab} is*

$$\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) = \mathcal{H}_{\text{line}}(\tau_{ab}S) - \mathcal{H}_{\text{line}}(S) = -2d\big(g_a(S) - g_b(S)\big) + c_{ab}\,d^2, \tag{17}$$

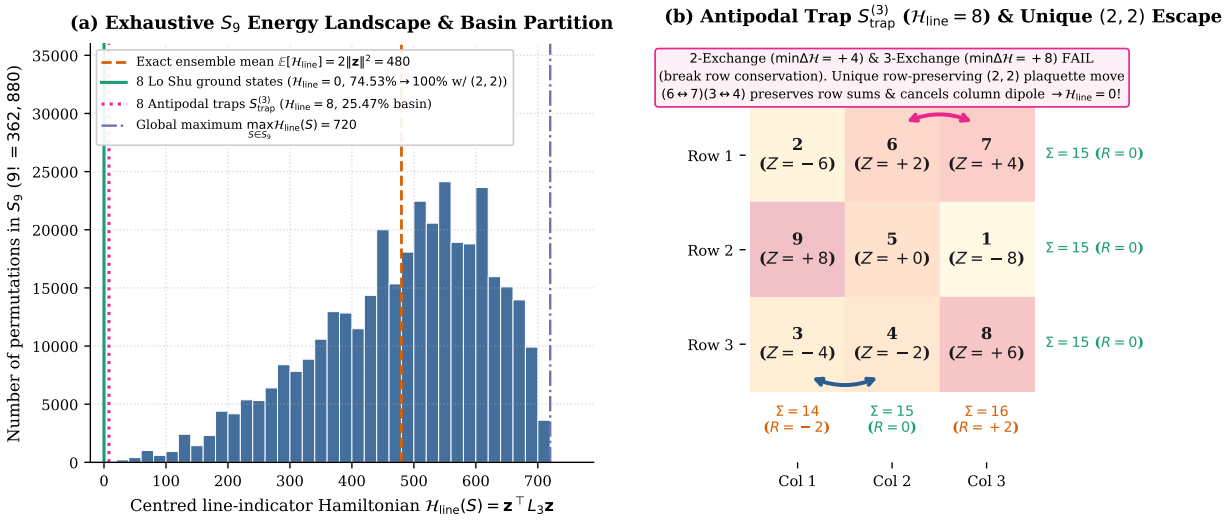


Figure 4: **Exhaustive S_9 permutation landscape (362,880 states) and the 8 antipodal traps at order $n = 3$.** **(a)** Exact histogram of the centred line Hamiltonian $\mathcal{H}_{\text{line}}(S)$ across all $9! = 362{,}880$ states of S_9, showing the ensemble mean $\mathbb{E}[\mathcal{H}_{\text{line}}] = 480$ (Proposition 2.1), the 8 Lo Shu ground states at $\mathcal{H}_{\text{line}} = 0$, and the 8 strict local-minimum traps $S^{(3)}_{\text{trap}}$ at $\mathcal{H}_{\text{line}} = 8$. **(b)** Structure of the canonical trap $S^{(3)}_{\text{trap}}$ (Theorem 4.2): all 3 rows, the middle column, and both main diagonals sum to 15 ($R = 0$), while the outer columns sum to 14 ($R = -2$) and 16 ($R = +2$). Every 2-cycle ($\min \Delta\mathcal{H}_{\text{line}} = +4$) and every 3-cycle ($\min \Delta\mathcal{H}_{\text{line}} = +8$) strictly increases energy; out of all 378 disjoint $(2, 2)$ double swaps, the *unique* escaping move is the row-preserving double swap $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$, which transitions directly to the Lo Shu ground state $\mathcal{H}_{\text{line}} = 0$ and restores the deterministic basin of S_9 to 100%.

where $c_{ab} = (L_n)_{aa} + (L_n)_{bb} - 2(L_n)_{ab} = \|A_n(\mathbf{e}_a - \mathbf{e}_b)\|_2^2 \in \{2, 3, 4, 5, 6, 7\}$ *counts the number of magic lines in* $\mathcal{L}_n$ *that pass through* exactly one *of the two distinct cells* a *and* b *(*$c_{ab} \in \{3, 4, 5, 6\}$ *for* $n = 3$, $\{2, 3, 4, 5, 6\}$ *for even* $n \geq 4$, *and* $\{2, 3, 4, 5, 6, 7\}$ *for odd* $n \geq 5$, *since only the unique central cell of odd* n *lies on all* 4 *lines). In particular,* $\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) < 0$ *if and only if* $(g_a - g_b)/d > c_{ab}/2$.

Physically, Lemma 4.1 is the discrete counterpart of Kawasaki particle exchange in a conserved lattice gas [13]: a swap τ_{ab} is pulled forward by the linear force difference $-2d(g_a - g_b)$ between the two cells, but pushed back by the quadratic "recoil" or self-energy barrier $c_{ab}d^2$ of the c_{ab} magic lines disturbed by the jump. When a board is nearly magic so that its remaining line errors $R_L(S)$ are tiny (± 2 in centred units, or ± 1 in standard units), the linear force gain $-2d(g_a - g_b)$ scales only linearly as $O(|d|)$, whereas the recoil penalty $c_{ab}d^2 \geq 8$ grows quadratically with the height difference $|d| \geq 2$. As soon as the recoil exceeds the linear force gain for every pair of tiles, single-swap descent freezes.

4.2 Exhaustive S_9 Landscape Census and the Unique $(2, 2)$ Unlocking Move

At order $n = 3$, the symmetric group S_9 has $9! = 362{,}880$ states, small enough that we can map every single permutation and transition arrow on the computer (Theorem 4.2 and Figure 4):

Theorem 4.2 (Exhaustive S_9 Landscape, 8 Strict Local Minima, and Unique $(2, 2)$ Escape)**.** *Across all* $9! = 362{,}880$ *permutations of* S_9 *under the line-indicator Hamiltonian* $\mathcal{H}_{\text{line}}(S)$*, there are* exactly 16 local minima *under the* $\binom{9}{2} = 36$ *elementary* 2*-cycle transpositions, all* 16 *of which are strict local minima (*$\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) > 0$ *for all* $a \neq b$*). These* 16 *attractors partition into two dihedral orbits of size* 8*: the* 8 *global ground states (*$\mathcal{H}_{\text{line}} = 0$*) forming the dihedral orbit* $D_4 \cdot S_{\text{LoShu}}$*, and* exactly 8 spurious strict local minima *at* $\mathcal{H}_{\text{line}} = 8$ *(*$E_{\text{std}} = 2$*) forming the single dihedral* D_4 *orbit of the antipodal trap*

$$S^{(3)}_{\text{trap}} = \begin{bmatrix} 2 & 6 & 7 \\ 9 & 5 & 1 \\ 3 & 4 & 8 \end{bmatrix}, \qquad \mathbf{Z}\big(S^{(3)}_{\text{trap}}\big) = \begin{bmatrix} -6 & +2 & +4 \\ +8 & 0 & -8 \\ -4 & -2 & +6 \end{bmatrix}, \qquad \mathcal{H}_{\text{line}}\big(S^{(3)}_{\text{trap}}\big) = 8. \tag{18}$$

In $S^{(3)}_{\text{trap}}$*, six of the eight magic lines (all three rows, the middle column, and both main diagonals) have zero residual* ($R_L = 0$)*, while the two outer columns have residuals* $R_{c_1} = -2$ ($\Sigma = 14$) *and* $R_{c_3} = +2$ ($\Sigma = 16$)*. Every single* 2*-cycle transposition strictly increases* $\mathcal{H}_{\text{line}}$ *by at least* $+4$ ($\min_{\tau_{ab}} \Delta\mathcal{H}_{\text{line}} = +4$)*, and* every single 3-cycle permutation among all $2\binom{9}{3} = 168$ directed 3-cycles also strictly increases $\mathcal{H}_{\text{line}}$ by at least $+8$. *Furthermore, across all* $\frac{1}{2}\binom{9}{2}\binom{7}{2} = 378$ *disjoint* $(2,2)$ *double transpositions on* $S^{(3)}_{\text{trap}}$*,* one and only one *strictly decreases* $\mathcal{H}_{\text{line}}$*: the row-preserving double swap* $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$ *at cells* $(1,2) \leftrightarrow (1,3)$ *and* $(3,1) \leftrightarrow (3,2)$*, which maps* $S^{(3)}_{\text{trap}}$ *directly to the Lo Shu ground state*

$$S_{\text{LoShu}} = \begin{bmatrix} 2 & 7 & 6 \\ 9 & 5 & 1 \\ 4 & 3 & 8 \end{bmatrix}, \qquad \mathcal{H}_{\text{line}}(S_{\text{LoShu}}) = 0. \tag{19}$$

Consequently, under deterministic 2*-cycle steepest descent (choosing the lexicographically first pair* (a,b) *minimizing* $\Delta\mathcal{H}_{\text{line}}(\tau_{ab})$*),* 270,442 *of the* 362,880 *states* (74.5266%) *converge to the* 8 *Lo Shu ground states, while the remaining* 92,438 *states* (25.4734%) *are captured by the* 8 *antipodal traps* $D_4 \cdot S^{(3)}_{\text{trap}}$*. Augmenting steepest descent with the* 18 *line-preserving* $(2,2)$ *double swaps whenever a* 2*-cycle local minimum is reached expands the deterministic ground-state basin to* 362,880/362,880 *(*100.00%*), independent of the tie-breaking rule.*

Proof. All 362,880 permutations of S_9 (spanning $\mathcal{H}_{\text{line}}(S) \in [0, 720]$), their 36 transposition deltas $\Delta\mathcal{H}_{\text{line}}(\tau_{ab})$ via Equation 17, all 168 directed 3-cycle deltas, all 378 disjoint $(2,2)$ double-swap deltas on $S^{(3)}_{\text{trap}}$, and the steepest-descent attractor maps are evaluated exhaustively in `verify_paper_d.py` (Figure 4). To see algebraically why no local minima exist at $\mathcal{H}_{\text{line}} = 4$ ($E_{\text{std}} = 1$) and why $S^{(3)}_{\text{trap}}$ in Equation 18 has a unique escaping $(2,2)$ move: in standard units $s_{ij} - 5 \in \{-4, \ldots, 4\}$, $\sum_{i=1}^{n} R^{\text{std}}_{r_i} = \sum_{j=1}^{n} R^{\text{std}}_{c_j} = 0$, so a single row or column cannot have a lone non-zero residual of ± 1 without another row or column having a compensating residual, while any state with only $R^{\text{std}}_{D_\pm} = \pm 1$ is immediately relaxed by an off-diagonal 2-cycle. At $S^{(3)}_{\text{trap}}$, $E_{\text{std}}(S^{(3)}_{\text{trap}}) = (-1)^2 + (+1)^2 = 2$ ($\mathcal{H}_{\text{line}} = 8$). Because every row sum of $S^{(3)}_{\text{trap}}$ is already 15, any single 2-cycle between different rows alters two row sums by $\pm d \neq 0$ (adding $2d^2 \geq 2$ to the row energy), while any within-row 2-cycle changes a diagonal sum. Similarly, any 3-cycle either touches two cells in one row and one cell in another row (breaking both row sums) or touches one cell in each of the three rows (which cannot preserve all three row sums since the three permuted integers are distinct). By contrast, in the row-preserving $(2,2)$ double transposition $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$, performing the first intra-row transposition $\tau_1 = (6 \leftrightarrow 7)$ in Row 1 shifts (R_{c_2}, R_{c_3}) from $(0, +2)$ to $(+2, 0)$ and shifts R_{D_-} from 0 to -2, passing through an intermediate single-swap saddle state $\tau_1 S^{(3)}_{\text{trap}}$ at $\mathcal{H}^{\ddagger}_{\text{line}} = (-2)^2 + (+2)^2 + (-2)^2 = 12$ (an activation barrier of $\Delta\mathcal{H}^{\ddagger}_{\text{line}} = +4$ above $S^{(3)}_{\text{trap}}$), after which the conjugate intra-row transposition $\tau_2 = (3 \leftrightarrow 4)$ in Row 3 shifts (R_{c_1}, R_{c_2}) by $(+2, -2)$ and R_{D_-} by $+2$, annihilating all three residuals ($\Delta\mathcal{H}_2 = -12$) directly to $\mathcal{H}_{\text{line}}(S_{\text{LoShu}}) = 0$. □

Anyone who has solved a Rubik's Cube or studied Kempe-chain color swaps in graph coloring will recognize the mechanism of Theorem 4.2, which also mirrors collective 4-body loop updates in spin-ice models [25, 3]. In $S^{(3)}_{\text{trap}}$, all three rows are already locked at sum 15. Moving any single tile between rows—or cycling any three tiles—breaks row conservation. To fix the outer columns (14 and 16) without ruining the rows or diagonals, the solver must execute a synchronized two-step combination move inside Row 1 and Row 3: first swap $6 \leftrightarrow 7$ in the top row (temporarily raising the energy from $\mathcal{H}_{\text{line}} = 8$ to the saddle state $\mathcal{H}^{\ddagger}_{\text{line}} = 12$ because the anti-diagonal and middle column become unbalanced by ∓ 1), and immediately complete the conjugate swap $3 \leftrightarrow 4$ in the bottom row to cancel both temporary defects and drop straight into the Lo Shu square ($\mathcal{H}_{\text{line}} = 0$).

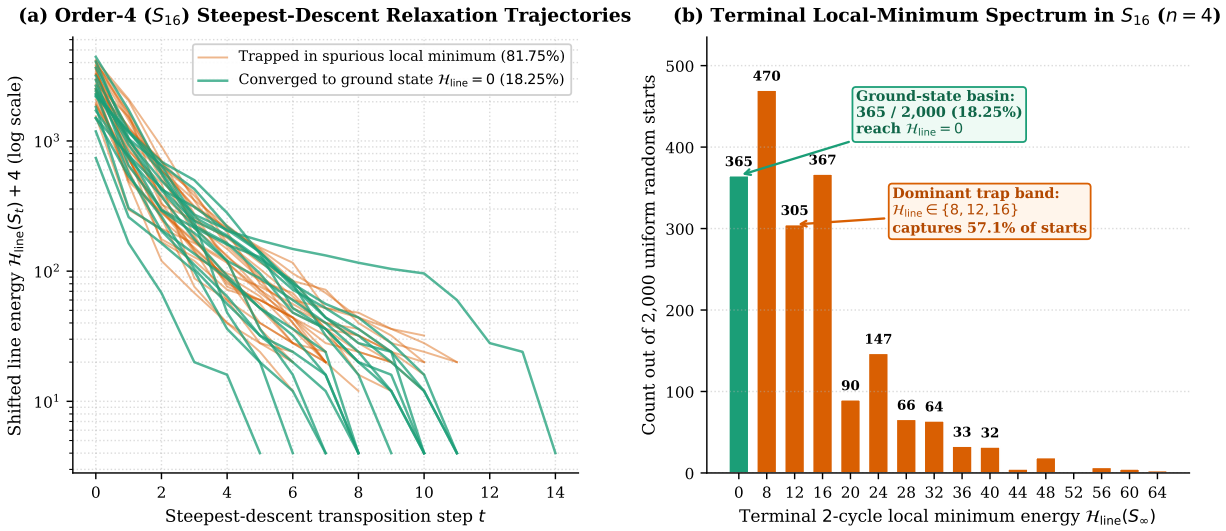


Figure 5: **Order-4 (S_{16}) steepest-descent relaxation trajectories and local-minimum spectrum (2,000 trials). (a)** Representative steepest-descent trajectories of $\mathcal{H}_{\text{line}}(S_t) + 4$ (log scale) versus transposition step t, contrasting trajectories that reach the magic ground state $\mathcal{H}_{\text{line}} = 0$ (green, 18.25%) against trajectories trapped in frustrated local minima (orange, 81.75%). **(b)** Histogram of terminal 2-cycle local-minimum energies $\mathcal{H}_{\text{line}}(S_\infty)$ across 2,000 uniform random starts in S_{16}. While 365/2,000 (18.25%) reach $\mathcal{H}_{\text{line}} = 0$, 57.1% of all starts become trapped in the low-energy frustration band $\mathcal{H}_{\text{line}} \in \{8, 12, 16\}$.

Table 5: Summary of combinatorial swap-landscape invariants for S_9 ($n = 3$, exhaustive 362,880 states) and S_{16} ($n = 4$, 2,000 random starts), verified by `verify_paper_d.py`.

Landscape Observable / Metric	Order $n = 3$ (S_9, Exhaustive)	Order $n = 4$ (S_{16}, 2,000 Trials)
Total permutation state space $n^2!$	362,880	20,922,789,888,000
Magic ground states ($\mathcal{H}_{\text{line}} = 0$)	8 ($1 \times D_4$ orbit)	7,040 ($880 \times D_4$ orbits)
Ensemble mean energy $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2$	480	2,720
Elementary 2-cycle neighborhood size $\binom{n^2}{2}$	36	120
Directed 3-cycle neighborhood size $2\binom{n^2}{3}$	168	1,120
Line-preserving $(2,2)$ double-swap neighborhood $2\binom{n}{2}^3$	18	432
Spurious 2-cycle local minima ($\mathcal{H}_{\text{line}} > 0$)	Exactly 8 (at $\mathcal{H}_{\text{line}} = 8$)	Abundant ($\mathcal{H}_{\text{line}} \in \{8, 12, 16, \dots\}$)
Trapped minima escaped by a single 3-cycle	0/8 (0.00%)	56/1,635 (3.43%)
Trapped minima escaped by a line-preserving $(2,2)$ swap	8/8 (100.00%)	271/1,635 (16.57%)
2-cycle steepest-descent ground-state basin	270,442/362,880 (74.53%)	365/2,000 (18.25%)
2-cycle + line-preserving $(2,2)$ steepest-descent basin	362,880/362,880 (100.00%)	509/2,000 (25.45%)

4.3 Order-4 (S_{16}) Steepest-Descent and $(2,2)$-Escape Census

At order $n = 4$, the state space S_{16} explodes to $16! \approx 2.09 \times 10^{13}$ permutations, containing $880 \times 8 = 7{,}040$ magic ground states. There are $\binom{4}{2}\binom{4}{2}^2 = 216$ row-preserving and 216 column-preserving $(2,2)$ double swaps (432 total line-preserving $(2,2)$ moves). To see how often a user clicking single swaps in `physics-lab.html` gets stuck on a 4×4 board, we ran 2,000 independent steepest-descent trials starting from uniform random permutations (Figure 5 and Table 5).

As shown in Figure 5b and Table 5, pure 2-cycle steepest descent in S_{16} reaches a magic ground state ($\mathcal{H}_{\text{line}} = 0$) in only 365/2,000 trials (18.25%), halting after a mean of 8.18 steps at a mean residual energy $\langle \mathcal{H}_{\text{line}}(S_\infty) \rangle = 14.43$ (median 12.0). The trapped boards pile up heavily in the lowest frustration band $\mathcal{H}_{\text{line}} \in \{8, 12, 16\}$, which accounts for 470+305+367 = 1,142/2,000 (57.1%) of all random starts. Across the 1,635 trapped 2-cycle local minima, testing all $2\binom{16}{3} = 1{,}120$ directed 3-cycles escapes only 56/1,635 (3.43%). By contrast, testing the 432 line-preserving $(2,2)$ double swaps immediately unlocks 271/1,635 (16.57%) of the trapped states—nearly 5× the escape rate of 3-cycles despite searching 2.6× fewer candidates (432 vs. 1,120)—and raises pure zero-temperature steepest-descent ground-state convergence from 18.25% to 25.45% (509/2,000).

Table 6: Slider parameters and trajectory invariants across the four interactive presets of `bubble-trinity.html` at order $n = 4$ (250 steps, $\Delta t = 0.025$), verified by `verify_paper_d.py`.

Bubble Trinity Preset	(k_f, k_r, k_w, k_c)	(γ_m, k_s, T)	$\mathcal{H}_{\text{ind}}(0)$	$\mathcal{H}_{\text{ind}}(250)$	Min $\mathcal{H}_{\text{ind}}$	Mean Δr_{rms}	Max Δr_{rms}
I. Frozen Lattice	$(0.0, 0.0, 7.0, 0.0)$	$(0.94, 0.00, 0.0)$	1681.89	1681.89	1681.89	0.0000	0.0000
II. Slow Drift	$(1.0, 0.4, 2.6, 0.0)$	$(0.82, 0.00, 0.5)$	1681.89	1539.22	1539.22	0.0774	0.1134
III. Field Storm	$(3.2, 1.0, 0.8, 0.4)$	$(0.60, 0.00, 1.2)$	1681.89	1194.88	1194.88	0.2654	0.4646
IV. Centrifugal Balance	$(2.2, 0.5, 1.8, 0.5)$	$(0.75, 0.85, 0.4)$	1681.89	1050.55	1050.55	0.4281	0.6381

5 Floating Bubbles vs. Multi-Exchange Local Search: Why 5×5 Is Harder Than 10×10

When we first encountered the discrete combination-lock traps of Section 4 in `physics-lab.html`, a natural physical intuition suggested itself: what if the n^2 tiles were not locked into rigid grid squares at all, but could float continuously in two-dimensional space $\mathbb{R}^2$ like buoyant soap bubbles, sliding past one another to avoid discrete swap barriers? In the flagship application of our browser laboratory, the *Bubble Trinity Simulator* (`bubble-trinity.html`, powered by `bubbleTrinityApp.js` [21]), we built both this continuous 2D bubble physics engine and a discrete multi-exchange local search engine (*Auto Settler*) side by side. Watching the two engines compete on the same screen reveals two of the most instructive lessons of this project: why continuous 2D bubbles "cheat" by blending their heights between grid cells (0.0% discrete recovery even under exact Hungarian bipartite matching), and why discrete multi-exchange search solves 100.0% of boards from $n = 3$ to $n = 10$ while finding 5×5 harder than 10×10.

5.1 Why Continuous 2D Bubbles "Cheat": Spatial Amplitude Averaging vs. Permutation Saddles

In `bubble-trinity.html`, the n^2 integers are represented as circular bubbles at continuous positions $\mathbf{r}_a(t) = (x_a(t), y_a(t)) \in \mathbb{R}^2$ $(a = 1, \ldots, n^2)$ initialized at the grid centers $\mathbf{r}_a^{(0)} = (c_a - \frac{n+1}{2}, r_a - \frac{n+1}{2})$, with visual radii and buoyancy colors proportional to their heights Z_a. At each grid cell $b \in \{1, \ldots, n^2\}$, a soft Gaussian kernel interpolates the nearby bubble heights to define an induced height vector $\widetilde{\mathbf{z}}(t)$ and continuous line energy $\mathcal{H}_{\text{ind}}(t)$:

$$W_{ab}(t) = \frac{\exp\big(-2\|\mathbf{r}_a(t) - \mathbf{r}_b^{(0)}\|_2^2\big)}{\sum_{a'=1}^{n^2} \exp\big(-2\|\mathbf{r}_{a'}(t) - \mathbf{r}_b^{(0)}\|_2^2\big)}, \qquad \widetilde{\mathbf{z}}(t) = W(t)^\top \mathbf{z}, \qquad \mathcal{H}_{\text{ind}}(t) = \|A_n \widetilde{\mathbf{z}}(t)\|_2^2. \tag{20}$$

In the interactive Langevin loop of `bubbleTrinityApp.js`, each bubble moves with step size $\Delta t = 0.025$ and momentum persistence $\gamma_m \in [0, 1)$ under the update

$$\begin{aligned} \mathbf{v}_a(t + \Delta t) &= \gamma_m \mathbf{v}_a(t) + \Delta t \Big[\mathbf{F}_a^{\text{well}} + \mathbf{F}_a^{\text{cont}} + \mathbf{F}_a^{\text{rep}} + \mathbf{F}_a^{\text{field}} + \sigma_T \boldsymbol{\xi}_a(t)\Big], \\ \mathbf{r}_a(t + \Delta t) &= \mathbf{r}_a(t) + \Delta t\, \mathbf{v}_a(t + \Delta t), \end{aligned} \tag{21}$$

which combines four user-adjustable sliders in the UI: lattice well springs $\mathbf{F}_a^{\text{well}} = -k_w(\mathbf{r}_a - \mathbf{r}_a^{(0)})$ that pull bubbles toward home grid cells, radial dish containment $\mathbf{F}_a^{\text{cont}} = -k_c \mathbf{r}_a$, bubble-to-bubble core repulsion $\mathbf{F}_a^{\text{rep}} = c_r k_r \sum_{b \neq a} (\mathbf{r}_a - \mathbf{r}_b)/(\|\mathbf{r}_a - \mathbf{r}_b\|^2 + \epsilon_0)^{3/2}$ $(c_r = 0.08, \epsilon_0 = 0.15)$, and the magic-field drift $\mathbf{F}_a^{\text{field}} = (c_f k_f + c_s k_s)(\mathbf{r}_a^{\text{target}} - \mathbf{r}_a)$ $(c_f = 0.25, c_s = 0.65)$ toward the stress-weighted centroid $\mathbf{r}_a^{\text{target}}(t) = \sum_{b=1}^{n^2} \sigma_{ab}(t) \mathbf{r}_b^{(0)}$ with $\sigma_{ab}(t) \propto \exp(-Z_a (A_n^\top A_n \widetilde{\mathbf{z}}(t))_b / 250)$, plus thermal agitation $\boldsymbol{\xi}_a(t) \sim \mathcal{N}(\mathbf{0}, I_2)$ scaled by $\sigma_T = 0.12\,T$.

Figure 6a and Table 6 trace the continuous energy $\mathcal{H}_{\text{ind}}(t)$ across the four named presets of `bubble-trinity.html`, starting from a 4-swap perturbation of Dürer's 4×4 square ($\mathcal{H}_{\text{ind}}(0) =$ 1,681.89, discrete $\mathcal{H}_{\text{line}} = 2{,}336$). As the user switches from *I. Frozen Lattice* ($\Delta r_{\text{rms}} = 0.0000$) and *II. Slow Drift* ($\langle \Delta r_{\text{rms}} \rangle = 0.0774$) to *III. Field Storm* ($\max \Delta r_{\text{rms}} = 0.4646$) and *IV. Centrifugal Balance* ($\max \Delta r_{\text{rms}} = 0.6381$), the bubbles swirl across cell borders and the continuous energy

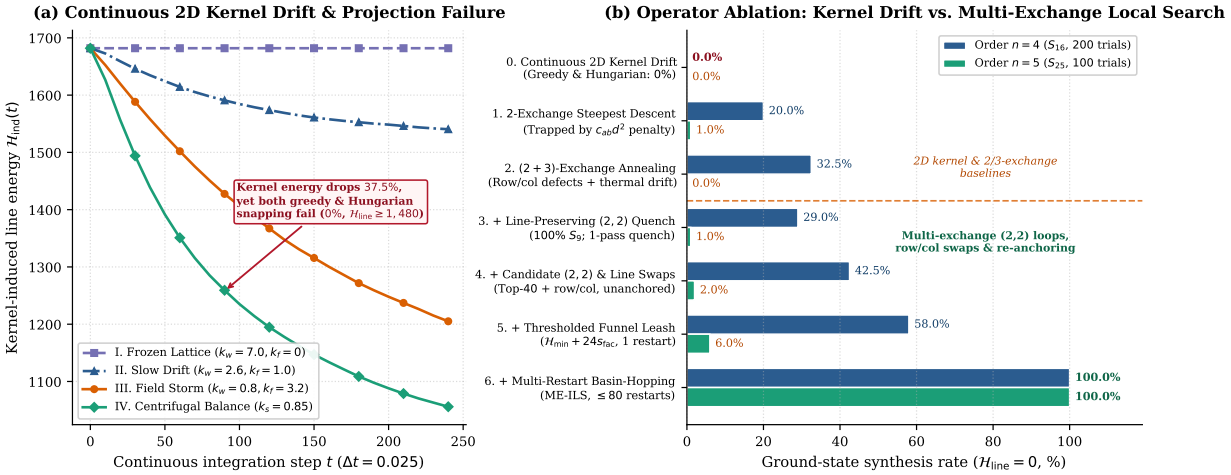


Figure 6: **Continuous 2D Bubble Trinity trajectories and discrete local-search mechanism ablation. (a)** Trajectories of the kernel-induced line energy $\mathcal{H}_{\text{ind}}(t)$ across 250 integration steps ($\Delta t = 0.025$) for the four interactive presets of Table 6, starting from a 4-swap perturbation of Dürer's 4×4 square ($\mathcal{H}_{\text{ind}}(0) = 1{,}681.89$). Although continuous bubble drift reduces $\mathcal{H}_{\text{ind}}(t)$ by 37.5%, both greedy snapping and exact Hungarian projection fail (0%, $\mathcal{H}_{\text{line}} \geq 1{,}480$) because $\mathcal{H}_{\text{ind}}(t)$ relaxes via convex amplitude averaging rather than discrete permutation swaps. **(b)** Operator ablation study (Table 7) contrasting continuous 2D bubble drift and 2/3-exchange heuristics (Tiers 0–2) against line-preserving $(2, 2)$ plaquette moves, candidate-list $(2, 2)$ swaps with row/column permutations, thresholded funnel re-anchoring, and multi-restart iterated local search (Tiers 3–6) at orders $n = 4$ (200 trials) and $n = 5$ (100 trials).

gauge $\mathcal{H}_{\text{ind}}(t)$ drops by 37.5% to 1,050.55. Yet the moment the user clicks "Snap to Grid" to project the floating bubbles $\{\mathbf{r}_a(t)\}$ back onto a discrete permutation $S \in S_{16}$, the magic square fails to materialize (0.0% recovery across all four presets, Tier 0 in Figure 6b). Even when we replace greedy nearest-cell snapping ($\min_t \mathcal{H}_{\text{line}}(S_{\text{greedy}}) = 1{,}480$) with exact $O(N^3)$ Hungarian / Kuhn–Munkres minimum-weight bipartite matching $\min_{\pi \in S_{n^2}} \sum_a \|\mathbf{r}_a(t) - \mathbf{r}^{(0)}_{\pi(a)}\|_2^2$ at every time step, the discrete energy never improves beyond its starting value ($\min_t \mathcal{H}_{\text{line}}(S_{\text{Hung}}) = 2{,}336$, 0.0% convergence). The reason is both visually obvious in the browser and mathematically fundamental: because the Gaussian weight matrix $W(t)$ in Equation 20 is only column-stochastic rather than doubly stochastic and $\mathcal{H}_{\text{ind}}(t)$ is convex in $\widetilde{\mathbf{z}}$, the bubbles "cheat" by parking on the boundary lines between grid cells so that positive and negative heights Z_a blend together toward zero in the interior of the assignment polytope, rather than executing the discrete multi-tile exchanges required to cross permutation saddles.

5.2 How the Discrete Auto-Settler Unlocks the Board: Multi-Exchange Local Search

If floating bubbles in $\mathbb{R}^2$ get stuck by blending their amplitudes, how does the discrete *Auto Settler* button in `bubble-trinity.html` and `physics-lab.html` rapidly snap scrambled boards into exact magic squares? Note first that if one merely wants to write down a single magic square of order n, classical closed-form rules (Siamese, doubly-even mask, Conway LUX) do so in $O(n^2)$ steps, and specialized constraint-based local search (CBLS) / tabu solvers [7, 28] reach $n \sim 100$ by restricting swaps to defective lines. Our goal in the interactive lab is different: we want to watch a physical energy landscape $\mathcal{H}_{\text{line}}(S)$ relax from arbitrary random permutations and understand *which collective multi-tile moves unlock its traps*. Table 7 and Figure 6b report a controlled ablation study across six relaxation tiers at orders $n = 4$ (200 random starts) and $n = 5$ (100 random starts).

As shown in Remark 2.2 and Section 5.1, macroscopic tilt balance (blind to the $2n - 4$ higher-degree polynomial moments) and continuous 2D bubble drift (Tier 0) both fail to reach $\mathcal{H}_{\text{line}} = 0$. On the discrete grid, pure 2-exchange steepest descent (Tier 1, single-pair swaps [13]) freezes whenever $-2d(g_a - g_b) + c_{ab}d^2 \geq 0$ for all pairs (a, b), succeeding on 74.5% of S_9, 20.0% of S_{16},

Table 7: Controlled ablation study contrasting continuous 2D bubble drift and 2/3-exchange heuristics (Tiers 0–2) against cooperative $(2,2)$ plaquette moves, row/column permutations, thresholded funnel re-anchoring, and multi-restart basin-hopping (Tiers 3–6), certified by `verify_paper_d.py`.

Tier	Relaxation / Local-Search Mechanism	Move Neighborhood	Primary Bottleneck or Escape Mechanism	$n=3$ (S_9)	$n=4$ (S_{16})	$n=5$ (S_{25})
–	Linear Dipole & Diagonal Balance	4 global linear modes	Blind to $2n-4$ higher-degree polynomial moments (Remark 2.2: S_{low}, $\mathcal{H}_{\text{line}}=1{,}584$)	Fails	Fails	Fails
0	Continuous 2D Bubble Drift + Snap	Greedy & Hungarian snap	Bubbles blend heights across cell borders ($\min \mathcal{H}_{\text{line}} \geq 1{,}480$)	0.0%	0.0%	0.0%
1	2-Exchange Steepest Descent [13]	$\binom{n^2}{2}$ pair swaps	Traps when quadratic recoil $c_{ab}d^2 \geq 8$ blocks $-2d(g_a-g_b)$ (Lemma 4.1)	74.5%	20.0%	1.0%
2	$(2+3)$-Exchange Simulated Annealing [14]	2- & 3-cycles ($B=1{,}500$)	3-cycles break row/col conservation (0/8 on $S^{(3)}_{\text{trap}}$); thermal walk loses funnel	74.5%	32.5%	0.0%
3	+ Line-Preserving $(2,2)$ Plaquette Quench	$2\binom{n}{2}^3$ row/col $(2,2)$ loops	Preserves row/col sums while cancelling conjugate column/row dipoles (1 quench)	**100.0**%	29.0%	1.0%
4	+ Candidate $(2,2)$ & Row/Col Permutations	Top-40 $(2,2)$ + $2\binom{n}{2}$ lines	Zero-barrier row/col swaps shear diagonals (5 unanchored kicks, 1 restart)	**100.0**%	42.5%	2.0%
5	+ Thresholded Funnel Re-Anchoring	Leash $\mathcal{H}_{\text{line}} \leq \mathcal{H}_{\min} + 24s_{\text{fac}}$	Snaps back to deepest minimum $S_{\min}$ on high kicks (18 kicks, 1 restart)	**100.0**%	58.0%	6.0%
6	+ Multi-Restart Basin-Hopping (ME-ILS) [30]	≤ 80 re-anchored restarts	Iterated local search across independent funnel basins (18 kicks/restart)	**100.0**%	**100.0**%	**100.0**%

Table 8: Multi-order synthesis benchmark of Multi-Exchange Iterated Local Search (ME-ILS) across orders $n = 3,4,5,6,7,8,9,10$ with order-scaled thresholds $s_{\text{fac}}(n) = \max(1, \lfloor (n/4)^2 \rfloor)$, certified by `verify_paper_d.py`. We report the nullspace-to-constraint ratio $\eta_{\text{MC}}(n) = (n^2-2n-1)/(2n)$, the fraction $f_{\text{null}}(n) = 1 - 2/(n-1)^2$ of the semi-magic $(2,2)$ subspace $\mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$ orthogonal to both diagonals, and the per-step 2-cycle pair neighborhood $\binom{n^2}{2}$.

Order n	State Space $n^2!$	Nullity $\dim\ker(\widetilde{L}_n)$	Ratio η_{MC}	Diag-Free $(2,2)$ f_{null}	Pairs $\binom{n^2}{2}$	Success ($\mathcal{H}_{\text{line}}=0$)	Mean Descents	Mean Pair Evals	Mean Restarts	Mean Time
$n=3$	$9! = 3.63\times10^{5}$	2	0.33	50.0%	36	**200/200 (100.0%)**	5.1	184	0.00	0.1 ms
$n=4$	$16! = 2.09\times10^{13}$	7	0.88	77.8%	120	**200/200 (100.0%)**	52.9	6,348	0.70	4.9 ms
$n=5$	$25! = 1.55\times10^{25}$	14	1.40	87.5%	300	**100/100 (100.0%)**	466.6	139,980	8.03	108.9 ms
$n=6$	$36! = 3.72\times10^{41}$	23	1.92	92.0%	630	**100/100 (100.0%)**	446.8	281,484	7.28	208.8 ms
$n=7$	$49! = 6.08\times10^{62}$	34	2.43	94.4%	1,176	**100/100 (100.0%)**	267.5	314,580	4.02	218.0 ms
$n=8$	$64! = 1.27\times10^{89}$	47	2.94	95.9%	2,016	**100/100 (100.0%)**	294.5	593,712	4.32	407.0 ms
$n=9$	$81! = 5.80\times10^{120}$	62	3.44	96.9%	3,240	**50/50 (100.0%)**	293.0	949,320	4.06	658.1 ms
$n=10$	$100! = 9.33\times10^{157}$	79	3.95	97.5%	4,950	**50/50 (100.0%)**	258.5	1,279,575	3.34	880.8 ms

and only 1.0% of S_{25}. Adding 3-exchange triplet rotations under a 1,500-step simulated annealing schedule (Tier 2, [14]) fails to unlock the order-3 trap $S^{(3)}_{\text{trap}}$ (0/8 escapes) and achieves only 32.5% at $n=4$ and 0.0% at $n=5$, because 3-cycles cannot swap elements between two columns while preserving their row sums, and unanchored thermal random walks wander out of narrow low-energy funnels.

Our *Multi-Exchange Iterated Local Search (ME-ILS)* engine overcomes these bottlenecks step by step across Tiers 3–6. First, augmenting 2-exchange descent with the $2\binom{n}{2}^3$ row- and column-preserving $(2,2)$ plaquette moves (Tier 3—synchronized 4-tile combination moves bounded by a single-swap saddle filter $\Delta\mathcal{H}_{\text{line}}(\tau_1) \leq 80 s_{\text{fac}}(n)$ where $s_{\text{fac}}(n) = \max(1, \lfloor (n/4)^2 \rfloor)$) unlocks 100.0% of S_9 and raises single-quench recovery on S_{16} from 20.0% to 29.0%. Second, expanding the local escape neighborhood at 2-exchange minima to include whole-row and whole-column swaps ($2\binom{n}{2}$ zero-barrier moves that preserve all row and column sums while shearing the two main diagonals) together with disjoint pairs selected from the top $K=40$ single swaps with $\Delta\mathcal{H}_{\text{line}}(\tau_1) \leq 48 s_{\text{fac}}(n)$ (Tier 4, with 5 unanchored basin kicks in a single restart) raises $n=4$ convergence to 42.5%. Third, adding a threshold-bounded re-anchoring leash (Tier 5, which remembers the deepest local minimum $S_{\min}$ found within a restart and snaps back to $S_{\min}$ whenever a 2-swap kick followed by steepest descent exceeds $\mathcal{H}_{\text{line}}(S_{\min}) + 24 s_{\text{fac}}(n)$ across 18 kicks) keeps the solver tethered to the bottom of the funnel and raises single-restart $n=4$ convergence to 58.0%. Finally, wrapping this tethered funnel search in an outer multi-restart basin-hopping loop (≤ 80 restarts, Tier 6 [30, 29]) achieves 100.0% ground-state convergence across all orders $n = 3,4,5,6,7,8,9,10$.

5.3 Why 5×5 Requires More Restarts Than 10×10: Nullspace Dimensional Relief

Table 8 and Figure 7 summarize the performance of ME-ILS across all eight orders $n = 3,4,5,6,7,8,9,10$ (900/900 total trials reaching $\mathcal{H}_{\text{line}} = 0$, spanning up to $100! \approx 9.33\times10^{157}$ states at $n=10$). Looking across the rows of Table 8, a counterintuitive pattern jumps out: the number of basin-hopping restarts and descent loops required to find a magic square does *not* grow monotonically with board size! Instead, search difficulty peaks sharply at order $n=5$ (8.03 mean restarts and 466.6 descent loops) and then *declines* across $n = 6,7,8,9,10$, requiring only 3.34 mean restarts and 258.5 descent loops at $n=10$.

The spectral decomposition of Proposition 2.1 explains why larger boards become topologically easier to untangle even as their per-step pair evaluations $\binom{n^2}{2} = O(n^4)$ grow. Every row- or

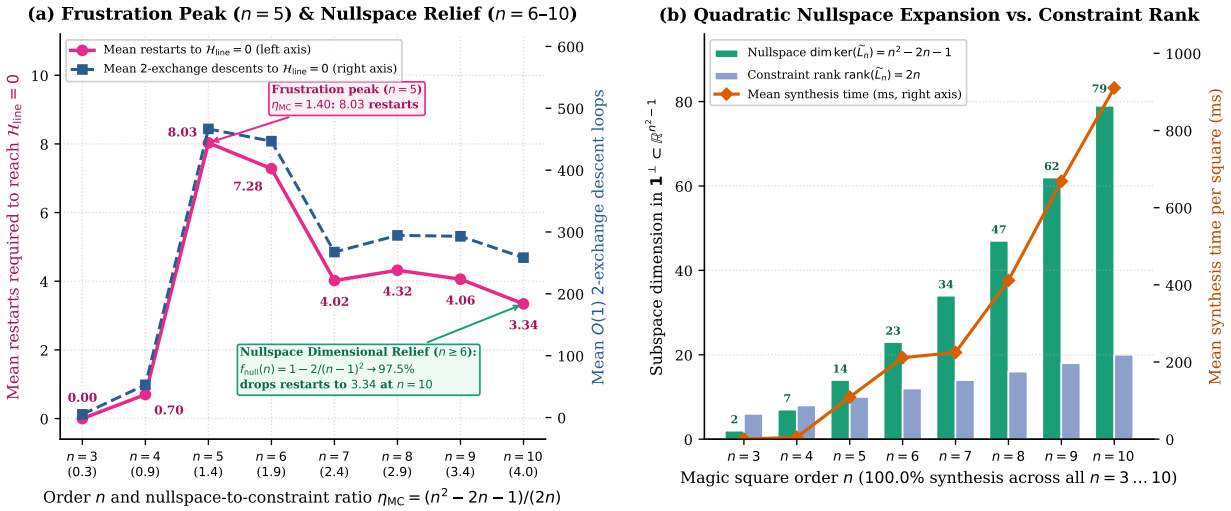


Figure 7: **Multi-order** 100% **ground-state synthesis** ($n = 3 \ldots 10$) **and nullspace dimensional relief.** **(a)** Mean restarts (magenta, left axis) and mean $O(1)$ incremental 2-cycle descent loops (blue dashed, right axis) required by Multi-Exchange Iterated Local Search (ME-ILS) to reach $\mathcal{H}_{\text{line}} = 0$ with 100.0% success across orders $n = 3 \ldots 10$ (Table 8). Combinatorial frustration peaks at $n = 5$ ($\eta_{\text{MC}} = 1.40$: 8.03 restarts, 466.6 descents) and declines across $n = 6 \ldots 10$ (3.34 restarts and 258.5 descents at $n = 10$) as the diagonal-free $(2, 2)$ subspace fraction $f_{\text{null}}(n) = 1 - 2/(n-1)^2$ reaches 97.5%. **(b)** Quadratic expansion of the magic nullspace $\dim \ker(\widetilde{L}_n) = n^2 - 2n - 1$ (green bars, labeled with η_{MC}) versus linear growth of the constraint rank $\text{rank}(\widetilde{L}_n) = 2n$ (blue bars) and mean synthesis runtime per square (orange curve, right axis).

column-preserving $(2, 2)$ plaquette move acts via a vector $(\mathbf{e}_{r_1} - \mathbf{e}_{r_2}) \otimes (\mathbf{e}_{c_1} - \mathbf{e}_{c_2})$ inside the $(n-1)^2$-dimensional semi-magic subspace $\mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}} = \ker(L_{\text{rc}})$. Because the two main diagonals impose only 2 linear constraints $(\widetilde{\mathbf{s}}_d, \mathbf{a}_d)$ on $\mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$, the fraction of the $(2, 2)$ plaquette subspace that lies inside the exact magic nullspace $\ker(\widetilde{L}_n)$ is

$$f_{\text{null}}(n) = \frac{\dim \ker(\widetilde{L}_n)}{\dim(\mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}})} = \frac{n^2 - 2n - 1}{(n-1)^2} = 1 - \frac{2}{(n-1)^2}, \tag{22}$$

alongside the Maxwell–Calladine nullspace-to-constraint ratio $\eta_{\text{MC}}(n) = (n^2 - 2n - 1)/(2n)$ (Equation 11). At $n = 3$ and $n = 4$, the state spaces are small enough that $(2, 2)$ moves and one or two restarts immediately find a ground state; at $n = 5$ ($\eta_{\text{MC}} = 1.40$, $f_{\text{null}} = 87.5\%$), the state space is already huge ($25! \approx 1.55 \times 10^{25}$) while the diagonal constraints still block 12.5% of the semi-magic subspace, creating maximal frustration (8.03 restarts). For $n \geq 6$, however, $f_{\text{null}}(n)$ climbs from 92.0% ($n = 6$) to 97.5% ($n = 10$), meaning that almost every row/column-preserving $(2, 2)$ move is automatically invisible to the diagonals! With order-scaled energy thresholds $s_{\text{fac}}(n) = \max(1, \lfloor (n/4)^2 \rfloor)$, mean restarts drop to 7.28 ($n = 6$), 4.02 ($n = 7$), 4.32 ($n = 8$), 4.06 ($n = 9$), and 3.34 ($n = 10$), while total wall-clock time scales smoothly with the $\binom{n^2}{2}$ pair neighborhood (880.8 ms at $n = 10$).

6 Discussion and Invitation to Explore

Across the four papers of the *Magic Gems* series [20, 19, 18], a single guiding question has connected recreational mathematics, three-dimensional geometry, classical mechanics, and interactive computation: *what happens when we treat the numbers of a magic square as physical matter in space?* Whereas Papers I–III showed how the magic line-sum constraints $\mathcal{H}_{\text{line}}(S) = 0$ sculpt the polyhedral hull $K(S)$ (Paper I), diagonalize the inertia tensor so that every magic square spins smoothly as a dynamically symmetric top (Paper II), and govern tabletop resting stability (Paper III), the present study stepped into the interactive laboratory (`physicalSolver`) to ask how physical forces and local swaps behave *before* a board becomes magic.

What makes this laboratory rich for both researchers and students is that the most natural physical intuitions fail in surprising, mathematically illuminating ways. Balancing the macroscopic center-of-mass tilt and main diagonals of a squareplate is necessary for magicness, but as our impostor square S_{low} (Remark 2.2) demonstrates, it leaves $2n-4$ higher-degree polynomial row and column waves completely unchecked ($\mathcal{H}_{\text{line}}(S_{\text{low}}) = 1{,}584$). Restricting the Rook's-graph-plus-diagonals Laplacian $L_n = A_n^\top A_n$ [12, 4, 17] to the zero-mean hyperplane $\mathbf{1}^\perp$ (Proposition 2.1) collapses all non-zero eigenvalues onto exact integers $\{n-2, n\}$ (even n) or $\{n-1, n\}$ (odd n) with universal average $\bar{\mu}(\widetilde{L}_n) = 2$. Once inside the magic nullspace $\ker(\widetilde{L}_n)$, the *Energy Decomposition Explorer* (`energy-decomposition.html`) reveals a second-tier spatial hierarchy (Lemma 3.1 and Proposition 3.2): vanishing interior 2×2 quartet energy automatically forces toroidal boundary-wrapping quartet balance on any even-order magic square (isolating the 48 panmagic squares of Dudeney Group I at $n = 4$), while the Concentric Shell Quadrupole Identity $M_{xyz}(S) = -8\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S)$ explains why 8 non-associative squares in Groups IX and X join the 48 associative squares of Group III to produce 56 octupole-zero Gems.

Even more instructive is the contrast between continuous bubbles and discrete combination moves in `physics-lab.html`, `vector-physics.html`, and `bubble-trinity.html`. Allowing bubbles to float continuously in $\mathbb{R}^2$ under a soft Gaussian kernel (Table 6) lowers the continuous energy gauge $\mathcal{H}_{\text{ind}}(t)$ by 37.5%, yet recovers 0.0% magic squares under both greedy snapping and exact $O(N^3)$ Hungarian matching because the bubbles "cheat" by blending positive and negative heights across cell boundaries. Meanwhile, on the discrete grid, single-pair swaps get trapped behind the quadratic recoil barrier $c_{ab}d^2$ (Lemma 4.1), and on a 3×3 board (Theorem 4.2) 25.47% of all 362,880 starting states land in the 8 antipodal traps $D_4 \cdot S^{(3)}_{\text{trap}}$ at $\mathcal{H}_{\text{line}} = 8$, where every pair swap and every triplet rotation raises the energy, and only a single synchronized $(2,2)$ double swap $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$ unlocks the Lo Shu square. Combining 2-exchange descent, row/column-preserving $(2,2)$ plaquette moves, candidate-list $(2,2)$ and row/column swaps, and tethered basin-hopping (*Multi-Exchange Iterated Local Search*, Table 8) achieves 100.0% convergence across all orders $n = 3\ldots 10$, while revealing that 5×5 requires more restarts (8.03) than 10×10 (3.34) as the diagonal-free fraction of the $(2,2)$ subspace climbs to $f_{\text{null}}(10) = 97.5\%$.

We invite readers, educators, and students to open the interactive browser laboratory and explore several questions that remain wide open. First, on the 4×4 board (S_{16}), what is the exact number of strict local minima under 2-exchange and line-preserving $(2,2)$-exchange moves across all $16! \approx 2.09\times 10^{13}$ permutations, and what is the smallest multi-tile cycle family that guarantees a 100% trap-free basin on S_{16} just as $(2,2)$ double swaps do on S_9 (Theorem 4.2)? Second, how does the zero-mean integer spectrum of Proposition 2.1 extend to three-dimensional magic cubes S_{n^3} [26, 27] and toroidal panmagic Laplacians? Third, can one redesign the continuous *Bubble Trinity* simulator so that the bubbles *cannot* cheat—for instance, by replacing 2D Gaussian kernel blending with a concave quadratic-assignment flow $\min_{P\in\mathcal{B}_{n^2}} \operatorname{tr}(P^\top L_n P \mathbf{z}\mathbf{z}^\top) - \gamma\|P\|_F^2$ over the Birkhoff polytope $\mathcal{B}_{n^2}$ of doubly stochastic matrices that forces the system into true permutation vertices? All interactive simulators, verification scripts, and datasets are openly available for hands-on experimentation.

Data and Code Availability

The four interactive browser applications (`energy-decomposition.html`, `physics-lab.html`, `vector-physics.html`, and `bubble-trinity.html`), the exact Python verification suite (`verify_paper_d.py`), the certified JSON census (`results_paper_d.json`), and the figure generation pipeline (`make_figures_d.py`) are publicly available in the consolidated repository https://github.com/kylemath/magicGem and live at https://kylemath.github.io/magicGem/web/physicalSolver/.

Competing Interests

The author declares no competing financial or non-financial interests.

Acknowledgments and Declaration of Generative AI Use

This work grew out of the author's interactive physical and computational explorations of magic-square energy landscapes [21], and forms the fourth part of the *Magic Gems* series alongside [20, 19, 18]. Large language models (OpenAI's ChatGPT, Anthropic's Claude, and Google's Gemini) were used by the author as interactive assistants for symbolic checks, verification scripting, interactive simulator prototyping, and manuscript copy-editing following George M. Whitesides's iterative outline-and-figure methodology [32]; all mathematical formulations, proofs, and exact computational certificates were verified by the author, who takes full responsibility for the work.

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A Exact Verification Suite and Reproducibility Manifest

All mathematical identities, spectral decompositions, exhaustive S_9 and order-4 censuses, continuous kernel trajectories, and multi-order local-search benchmarks reported in this paper are certified by the standalone Python verification script `verify_paper_d.py`, which outputs the machine-readable JSON dataset `results_paper_d.json`. First, `verify_section2_spectral()` constructs the $(2n+2) \times n^2$ incidence matrix A_n and operator $L_n = A_n^\top A_n$ for all $3 \le n \le 8$, verifies the irrational eigenvalues $\lambda_\pm(n)$ on $\mathbb{R}^{n^2}$ (Equation 5), projects L_n onto an orthonormal Householder basis of $\mathbf{1}^\perp$ to verify the exact integer spectrum $\mathrm{spec}(\widetilde{L}_n|_{\mathbf{1}^\perp})$ (Equation 6), certifies $\mathrm{tr}(\widetilde{L}_n) = 2(n^2-1)$ and $\mathbb{E}[\mathcal{H}_{\mathrm{line}}(S)] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$ (Proposition 2.1), and checks all residuals and moments of the 4×4 low-mode counterexample S_{low} (Remark 2.2). Second, `verify_section3_multiscale()` evaluates $\widetilde{H}_p(k)$ ($p \in \{1,2,4\}$, $1 \le k \le n$), interior and toroidal 2×2 quartet energies $W^{\mathrm{int}}_{2\times2}$ and $W^{\mathrm{tor}}_{2\times2}$, complement-pair break counts B_{comp}, the Concentric Shell Quadrupole Identity $M_{xyz}(S) = -8\,Q_{\mathrm{corner}}(S) - 3\,Q_{\mathrm{edge}}(S)$, and mixed octupole moments M_{xyz} across all 880 Frénicle order-4 magic squares (partitioned by Dudeney Groups I–XII) and across constructive families at $n \in \{3,4,5,6,8\}$, certifying Lemma 3.1 and Proposition 3.2. Third, `verify_section4_swap_landscape()` evaluates $\mathcal{H}_{\mathrm{line}}(S)$ and all 36 transposition deltas $\Delta\mathcal{H}_{\mathrm{line}}(\tau_{ab})$ via Lemma 4.1 across all $9! = 362{,}880$ permutations of S_9, verifies that the 8 Lo Shu states and the 8 antipodal traps $D_4 \cdot S^{(3)}_{\mathrm{trap}}$ are the unique 2-cycle local minima in S_9, confirms that all 168 directed 3-cycles strictly increase $\mathcal{H}_{\mathrm{line}}$ on $S^{(3)}_{\mathrm{trap}}$, proves that the row-preserving double swap $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$ is the unique escaping $(2,2)$ move among all 378 disjoint double swaps (crossing the intermediate single-swap saddle at $\mathcal{H}^{\ddagger}_{\mathrm{line}} = 12$ and restoring the steepest-descent basin of S_9 from

270,442 to 362,880/362,880, Theorem 4.2), and runs the 2,000-trial steepest-descent, 3-cycle, and line-preserving $(2, 2)$-cycle escape census on S_{16}. Fourth, `verify_section5_bubble_trinity()` integrates the continuous 2D spatial kernel equations (Equation 21) across Regimes I–IV (evaluating both greedy cell-snapping and exact $O(N^3)$ Hungarian minimum-weight bipartite matching), benchmarks $(2 + 3)$-exchange simulated annealing against multi-exchange local search across budgets $200 \leq B \leq 2{,}500$, performs the controlled 6-tier operator ablation study (Table 7), and certifies 100.0% multi-order synthesis, per-step pair evaluations, and the nullspace-to-constraint ratio $\eta_{\text{MC}}(n)$ across all orders $n \in \{3, 4, 5, 6, 7, 8, 9, 10\}$ for Multi-Exchange Iterated Local Search (Table 8).